Space Exploration & Aerospace Engineering

Part VI

Military, Intelligence and Space Security

Space is a strategic domain. This Part describes, using openly published information, how signals intelligence and missile-warning satellites work, and how radars and telescopes track tens of thousands of objects to keep orbits safe from collisions and debris.



Chapters 19–21 · pages 343–395 of the complete manual
First Edition · Current to October 2026 · Written in plain language with deep technical detail

ENGINEERING REFERENCE MANUAL

Space Exploration & Aerospace Engineering

A complete, illustrated guide to the space ecosystem — from stratospheric balloons to interstellar probes — explaining how every major class of space system works, how it is engineered, and where artificial intelligence is changing it.

First Edition · Information current to October 2026
A4 format · Fourteen Parts · Fifty-three chapters · Ten appendices
Version 1.0 · October 2026

About this edition

Space Exploration & Aerospace Engineering — An Engineering Reference Manual. First Edition, Version 1.0, October 2026.

This manual was written to be read by students, engineers, analysts, managers and curious readers. Each topic starts with plain-language explanations and builds up to the equations, numbers and design trade-offs that practising engineers use. Sentences are kept short and every technical term is explained in plain words when it first appears; the glossary (Appendix H) and the list of acronyms (Appendix G) collect them. Facts, mission status and programme milestones are current to October 2026; space programmes change quickly, so dates of future events should be read as plans, not certainties.

Figures, photographs and data

Diagrams, charts and graphs were drawn specifically for this manual from published physical relationships and public data. Photographs are from the NASA Image and Video Library and are used under NASA's media usage guidelines (NASA material is generally not subject to copyright in the United States); individual credits appear with each image and in Appendix J. NASA does not endorse this publication. Numerical values (masses, power levels, performance) are representative published figures and may differ slightly between sources and mission phases.

Safety and responsibility

This is an educational reference. It is not a design standard, a flight rule, or a substitute for the requirements, analyses and verification processes of a qualified engineering organisation. Chapters dealing with military and intelligence systems use only openly published information and describe principles, not operational or classified details.

How numbers are written

SI units are used throughout, with common aerospace units (nautical miles, feet, pounds-force) shown where the industry uses them. Large numbers use thin spacing (e.g., 35 786 km). "≈" means approximately, "~" means order-of-magnitude. Dates follow the ISO format (YYYY-MM-DD) in tables.

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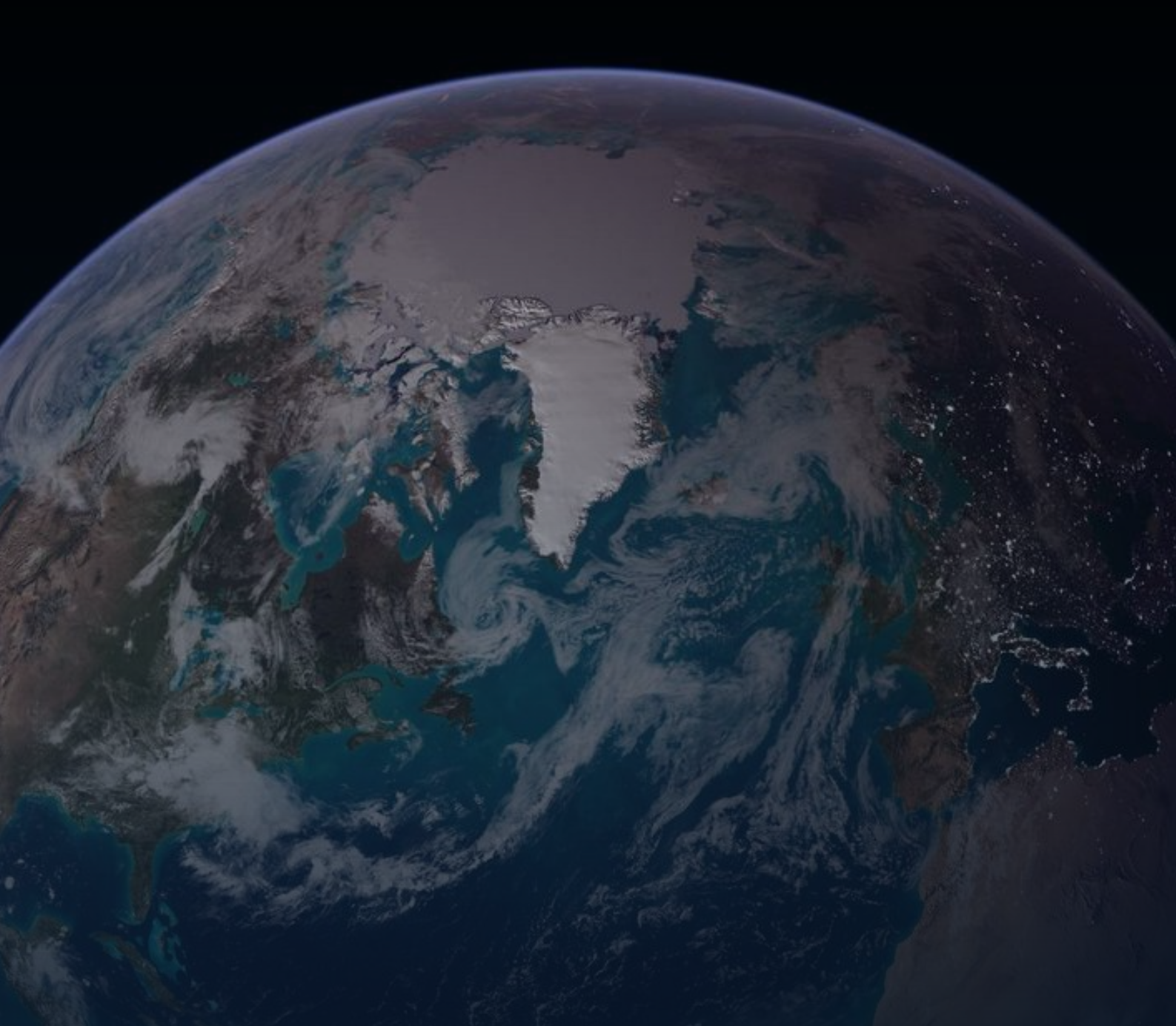
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This volume is one Part of *Space Exploration & Aerospace Engineering — An Engineering Reference Manual*. Page numbers, chapter numbers and figure numbers are the same as in the complete manual, so cross-references to other chapters (for example “Chapter 24”) and to the appendices refer to the complete book.



PART VI

Military, Intelligence and Space Security

Space is a strategic domain. This Part describes, using openly published information, how signals intelligence and missile-warning satellites work, and how radars and telescopes track tens of thousands of objects to keep orbits safe from collisions and debris.

19 National Security Space and Signals Intelligence

20 Missile Early Warning Satellites

21 Space Surveillance, Orbital Debris and Space Situational Awareness

Image: NASA/GSFC

CHAPTER

19 National Security Space and Signals Intelligence



Image: NASA/JSC

Space was a military and intelligence domain from the very beginning: the first reconnaissance satellites returned film from orbit in 1960, and today armed forces depend on space for intelligence, warning, navigation, communications and weather. This chapter describes, from public sources, the mission areas of national-security space and the organisations that conduct them; the principles of imagery and signals intelligence (SIGINT, ELINT and COMINT) from orbit, including the physics of intercepting and geolocating radio emitters; protected military satellite communications; the shift from a few exquisite satellites to proliferated constellations; the growing role of commercial space in conflict; and the counterspace threats — from anti-satellite missiles to jamming and cyber attacks — that have made space a contested domain, together with the legal and diplomatic efforts to limit them.

IN THIS CHAPTER

19.1 Space as a National Security Domain
19.2 Imagery Intelligence from Space
19.3 Signals Intelligence from Space
19.4 Military Satellite Communications
19.5 Military Weather, Launch and the Wider Enterprise
19.6 From Exquisite to Proliferated Architectures
19.7 Counterspace: A Contested Domain
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19.1 Space as a National Security Domain

19.1.1 Mission Areas

Military and intelligence organisations use space for the same reasons as civilians: global coverage, a vantage point above any territory, and the ability to relay signals over the horizon. But they have demands for resilience, security and timeliness that civil systems rarely face. National-security space is usually divided into the mission areas shown in Figure 19.1:

- **Intelligence, surveillance and reconnaissance (ISR)**: imagery (IMINT), signals intelligence (SIGINT), and measurement and signature intelligence (MASINT).
- **Missile warning and tracking**: infrared satellites that detect missile launches and nuclear detonations (Chapter 20).
- **Positioning, navigation and timing**: GPS and its military signals (Chapter 18).
- **Satellite communications (SATCOM)**: protected, wideband and narrowband links to forces worldwide.
- **Environmental monitoring**: military weather satellites.
- **Space domain awareness**: tracking and characterising objects and activities in space (Chapter 21).
- **Space control**: protecting one's own space systems and, if necessary, denying an adversary's use of space.
- **Launch and ranges**: assured access to space.

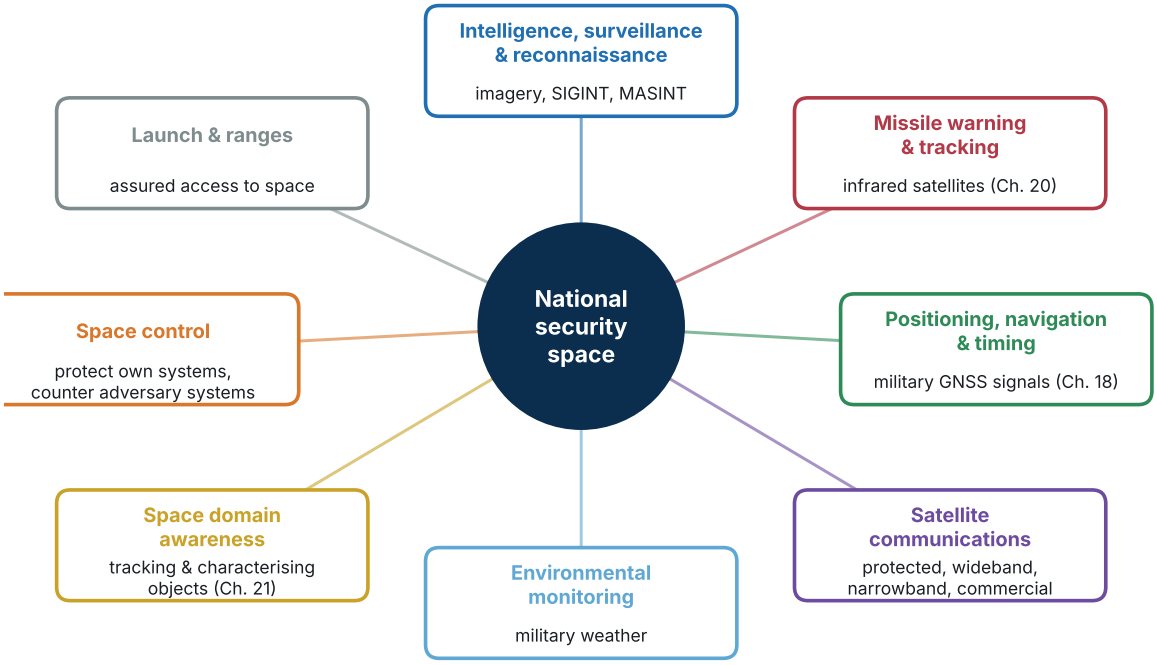


FIGURE 19.1 Mission areas of national-security space.

19.1.2 History

The first US reconnaissance satellites flew within three years of Sputnik (Figure 19.2). The Naval Research Laboratory's **GRAB** satellite, launched in June 1960 under cover of a scientific mission, intercepted Soviet air-defence radar signals — the world's first SIGINT satellite. Two months later, the **Corona** programme achieved the first recovery of exposed film from orbit, dropped in a capsule and caught in mid-air by an aircraft. A single early Corona mission photographed more of the Soviet Union than all previous U-2 flights combined, settling the debate about the "missile gap". Corona flew from 1960 to 1972, its resolution improving from about 12 m to under 2 m; its more than 800 000 images were declassified in 1995 and remain a valuable record of mid-20th-century landscapes. The Soviet Union's **Zenit** satellites, derived from the Vostok crewed spacecraft, followed from 1962.

Film-return systems gave way to electro-optical imaging satellites, which transmit images digitally, in the mid-1970s. The **Vela** satellites (from 1963) watched for nuclear tests in space and the atmosphere, and the **Defense Support Program** (from 1970) for ballistic-missile launches. The 1991 Gulf War, in which GPS, satellite communications, missile warning and reconnaissance supported coalition forces on an unprecedented scale, is often called the first "space war". The very existence of the US **National Reconnaissance Office (NRO)**, founded in 1961, was declassified only in 1992.

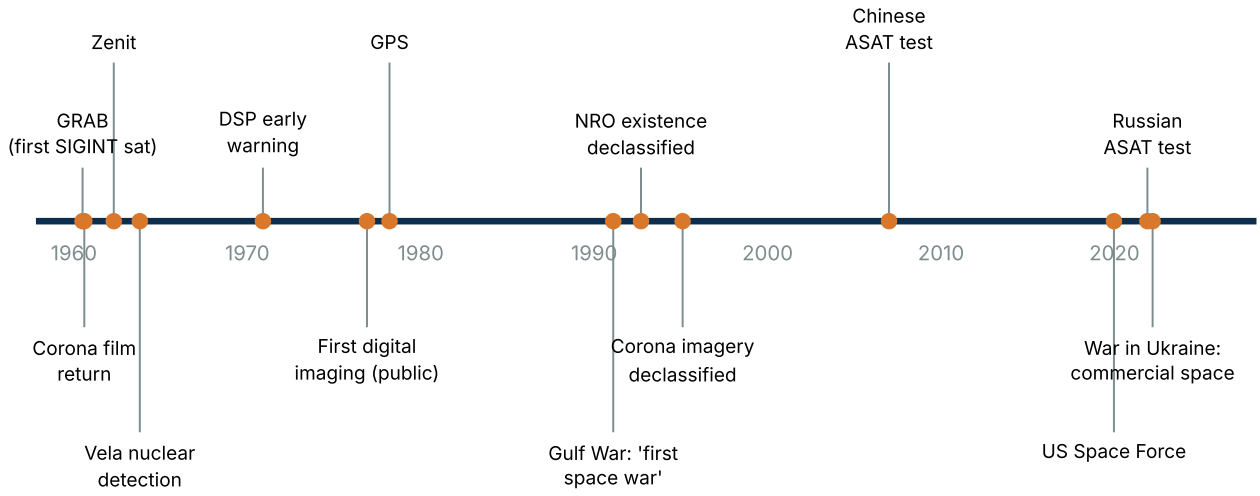


FIGURE 19.2 Milestones in national-security space.

19.1.3 Organisations

Many countries have reorganised their armed forces to treat space as a domain of operations alongside land, sea, air and cyberspace (Table 19.1). NATO recognised space as an operational domain in 2019.

TABLE 19.1 Selected military space organisations (public information).

Country	Military space organisation	Intelligence / other agencies	Notes
USA	US Space Force (2019); US Space Command (re-established 2019); Space Development Agency (2019)	National Reconnaissance Office; National Geospatial-Intelligence Agency	Largest military space budget; extensive use of commercial services
China	PLA Aerospace Force (2024, formed from the Strategic Support Force)	—	Yaogan, Gaofen, Shijian and TJS series; rapid growth
Russia	Aerospace Forces (VKS), Space Forces	GRU and others	Persona, Bars-M, Lotos-S/Liana, Tundra
France	Space Command (CDE, 2019)	DGSE, DRM	CSO imaging satellites; CERES SIGINT trio
United Kingdom	UK Space Command (2021)	—	Skynet SATCOM; partnerships and commercial data
Germany, Italy, Spain	National space commands	—	SARah, COSMO-SkyMed, PAZ radar satellites
Japan	Air and Space Self-Defense Force (renamed)	Cabinet Satellite Intelligence Center	Information Gathering Satellites (IGS)
India	Defence Space Agency (2019)	—	RISAT, EMISAT (ELINT), dedicated GSAT
Israel	Israeli Air Force / Ministry of Defence	—	Ofek imaging satellites

19.2 Imagery Intelligence from Space

Military imaging satellites use the same physics as civil Earth observation (Chapters 13–15): high-resolution optical and infrared telescopes, and synthetic aperture radars for all-weather, day-and-night coverage. The most capable systems are believed to use mirrors of about 2.4 m — the same size as the Hubble Space Telescope — giving resolutions of roughly 10–15 cm from low orbit (Chapter 14, Figure 14.3). Governments seldom publish the specifications of such systems. What distinguishes military imaging is less the sensor than the **system**: priority tasking, rapid processing, secure dissemination and the fusion of many sources into intelligence products.

19.2.1 Revisit and Constellation Design

For many military purposes the critical parameter is not resolution but **revisit** — how often a target can be observed. A single agile imaging satellite in a sun-synchronous orbit can view a mid-latitude target only about once every two to three days in daylight; doubling the number of satellites roughly halves the wait (Figure 19.3). Radar satellites, which can image at night as well, have about twice as many opportunities. To watch a target several times per day, a constellation of a dozen or more satellites is needed. An alternative is satellites in mid-inclination orbits, which pass over mid-latitudes more often and at varying local times. This is why both governments and commercial operators have moved towards larger constellations of smaller satellites.

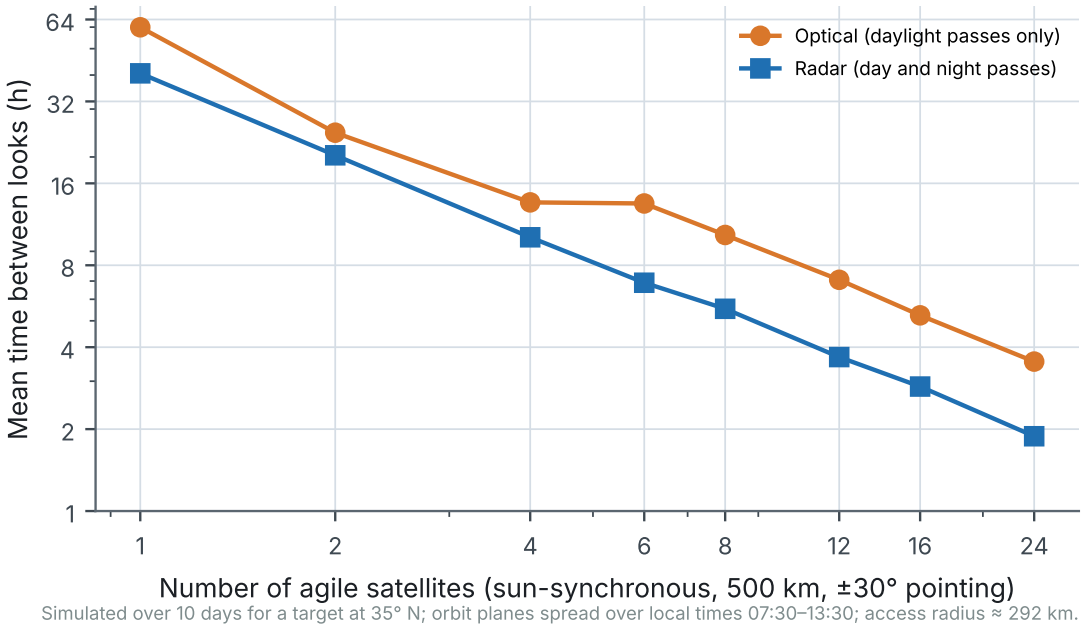


FIGURE 19.3 Simulated mean time between looks at a target at 35° N as a function of the number of agile imaging satellites.

TABLE 19.2 Selected government imaging and reconnaissance systems (public information).

Country	System (public name)	Type	Notes
USA	NRO electro-optical and radar satellites	Optical, SAR	Specifications classified; large-aperture optical systems since 1976
USA	NRO proliferated architecture	Optical and other sensors	Launched from 2024 with SpaceX and Northrop Grumman

Country	System (public name)	Type	Notes
France	CSO (Composante Spatiale Optique)	Optical	Three satellites (2018, 2020, 2025); shared with partners
Germany	SAR-Lupe, SARah	SAR	SARah constellation launched 2022–2023
Italy	COSMO-SkyMed (1st and 2nd generation)	SAR	Dual-use military and civil
Japan	Information Gathering Satellites (IGS)	Optical, SAR	Programme started after the 1998 North Korean missile overflight
Israel	Ofek, TecSAR	Optical, SAR	Launched westward from Israel on Shavit rockets
China	Yaogan, Gaofen series	Optical, SAR, ELINT	About 170 Yaogan satellites in orbit; some in triplets
Russia	Persona, Bars-M, Kondor	Optical, SAR	Smaller fleet than the USA or China
India	Cartosat, RISAT, EMISAT	Optical, SAR, ELINT	Dual-use national programmes

19.2.2 The Intelligence Cycle

Intelligence agencies describe their work as a cycle. It starts with **planning and direction** (what do decision-makers need to know?) and **collection** (tasking satellites and other sensors). Next comes **processing and exploitation** (turning raw data into information — locating, identifying and measuring objects). It ends with **analysis and production** (combining sources into assessments) and **dissemination**. Satellites contribute mainly to collection, but the bottlenecks have increasingly moved to processing and analysis. Modern constellations generate far more images than human analysts can examine. This is why AI-based object detection and change detection have become essential (Part XIII).

19.2.3 Commercial Imagery in National Security

Commercial imagery now complements government systems on a large scale. The USA buys imagery under the NRO's Electro-Optical Commercial Layer contracts and from commercial SAR providers, and many smaller nations rely entirely on commercial or allied data. During the war in Ukraine, governments have widely used commercial optical and radar imagery from companies such as Vantor (formerly Maxar), Planet, Airbus, ICEYE, Capella and BlackSky. The media have also published this imagery, transforming public understanding of military events. Commercial data are unclassified and shareable with allies and the public, which has given them diplomatic as well as operational value.

19.3 Signals Intelligence from Space

19.3.1 Disciplines

Signals intelligence (SIGINT) is the collection and analysis of electromagnetic signals emitted by an adversary's systems (Table 19.3). It is traditionally divided into:

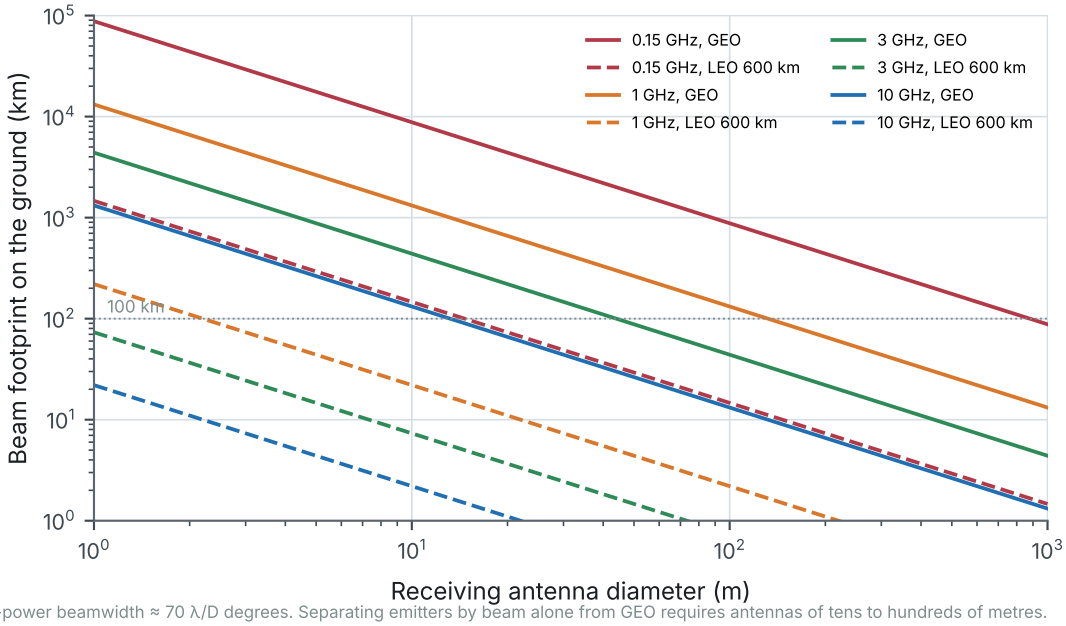
TABLE 19.3 Signals intelligence disciplines.

Discipline	Target	Typical frequencies	Information obtained
ELINT (electronic intelligence)	Radars and other non-communication emitters	≈ 0.1–40 GHz	Location, type and operating mode of radars; air-defence order of battle
COMINT (communications intelligence)	Voice, data and messaging	HF to Ka-band	Content (if decrypted), traffic patterns, location of communicators
FISINT (foreign instrumentation signals intelligence)	Telemetry from missiles, aircraft and spacecraft	VHF to S-band	Performance of weapons under test
Commercial RF geolocation	Ships with AIS switched off, satellite phones, push-to-talk radios, GNSS jammers	VHF to Ku-band	Location and track of emitters for maritime and security customers

19.3.2 Intercept Physics

A SIGINT satellite is a sensitive radio receiver with a large antenna. Intercepting radar signals is relatively easy because radars are powerful — often hundreds of kilowatts to megawatts of peak power — and even their side lobes are strong. Intercepting low-power communications from space is harder. The link budget (Chapter 7) shows that a 5 W VHF handheld radio received at 600 km arrives at about −95 dBm, which is easily detectable. At geostationary distance, however, it arrives about 36 dB weaker, so a significant receiving antenna is needed.

The bigger challenge is to **separate and locate** emitters. A receiving antenna's beamwidth is about $70\lambda/D$ degrees, so at low frequencies even large antennas have broad beams (Figure 19.4). From geostationary orbit, a 10 m antenna at 1 GHz has a beam about 1 300 km wide. At VHF, a beam narrow enough to isolate a single city would need an antenna several hundred metres across. Public reports describe very large deployable reflectors on some geostationary SIGINT satellites, and complementary geolocation techniques are used.



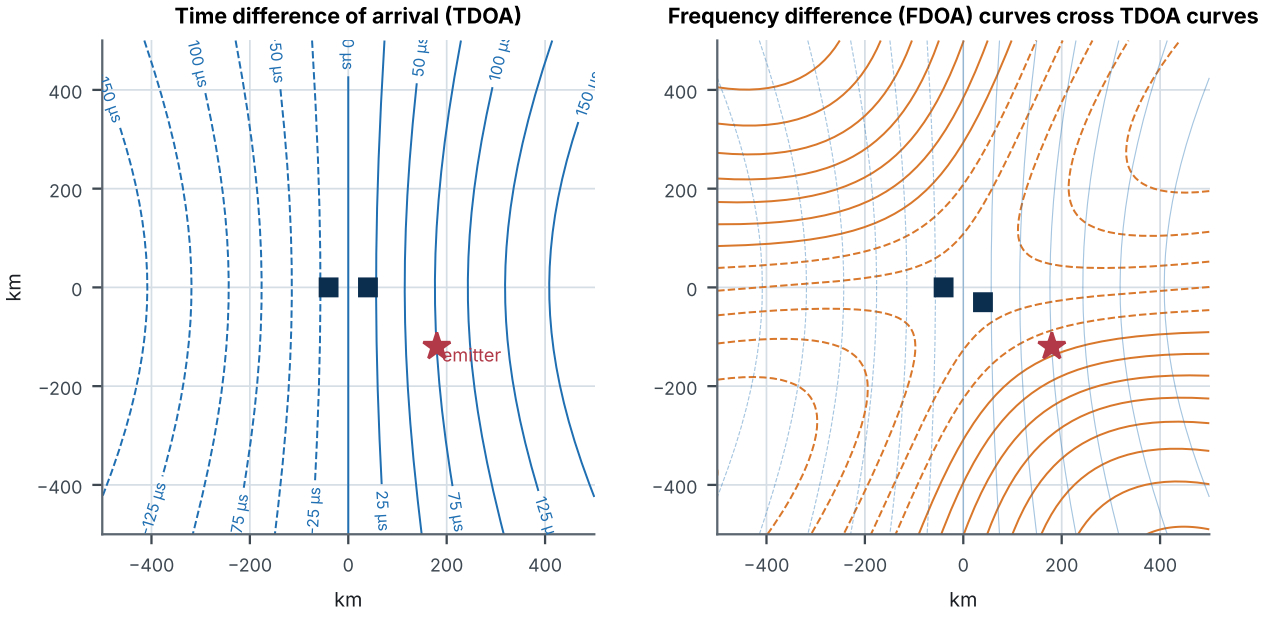
Half-power beamwidth ≈ 70 λ/D degrees. Separating emitters by beam alone from GEO requires antennas of tens to hundreds of metres.

FIGURE 19.4 Ground footprint of a receiving antenna beam as a function of antenna diameter, for four frequencies, from geostationary and low orbit. Isolating emitters by beam alone requires very large antennas, especially at low frequencies.

19.3.3 Geolocation Techniques

Emitters are located by measuring how their signals arrive at one or more receivers:

- **Angle of arrival (AOA)**: an interferometer measures the phase difference of a signal between two or more antennas separated by a baseline d; the phase difference $2\pi d \sin\theta/\lambda$ gives the arrival direction θ (Figure 19.6). Long baselines give precise but ambiguous angles; combinations of baselines resolve the ambiguity.
- **Time difference of arrival (TDOA)**: two satellites receive the same signal at slightly different times; each measured time difference places the emitter on a hyperbolic curve on the Earth's surface (Figure 19.5). Three satellites give two independent TDOAs whose curves intersect at the emitter.
- **Frequency difference of arrival (FDOA)**: because the satellites move, they receive slightly different Doppler shifts. The frequency difference defines another family of curves that cross the TDOA curves. So a pair of satellites measuring both can fix an emitter.
- **Single-satellite Doppler**: like the old Transit system in reverse, one satellite can locate a continuously transmitting emitter from the Doppler curve observed during a pass.



Two satellites 80 km apart at 600 km altitude (squares: sub-satellite points). Each measurement places the emitter on a curve; combining TDOA and FDOA (or three satellites) fixes its position.

FIGURE 19.5 Emitter geolocation from a pair of satellites. Lines of constant time difference of arrival (left) are hyperbolas on the ground; lines of constant frequency difference (right) cross them, so the two measurements together fix the emitter.

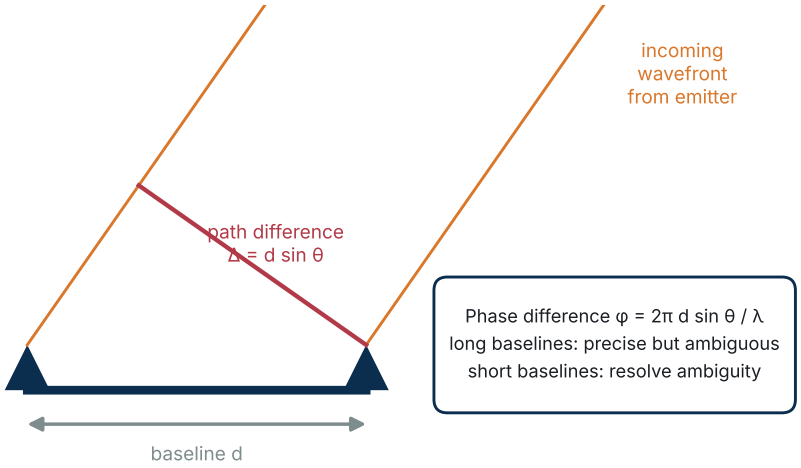


FIGURE 19.6 Angle-of-arrival measurement with a two-element interferometer.

These principles explain the distinctive configurations of SIGINT satellites reported in public sources. Ocean-surveillance systems such as the historic US "White Cloud"/NOSS series and China's Yaogan triplets fly **clusters** of two or three satellites tens of kilometres apart to measure TDOA from ship radars and communications. France's **CERES** system (three satellites launched in 2021) and its ELISA demonstrators locate radars for the French armed forces. Russia's Lotos-S/Liana system and India's EMISAT perform similar roles.

19.3.4 Commercial RF Geolocation

Since 2018, commercial companies have brought these techniques to an open market. **HawkEye 360** (USA) flies clusters of three microsatellites that detect and geolocate signals from marine VHF radios, AIS transponders, satellite phones, radars and GNSS jammers; by 2026 it had about 40 satellites in orbit. France's **Unseenlabs** locates ships from their radio emissions using single satellites, and Luxembourg's **Kleos Space** and others offer similar services. Their customers include coast guards fighting illegal fishing and smuggling — finding "dark" vessels that switch off their AIS transponders — and defence agencies.

19.3.5 MASINT

Measurement and signature intelligence (MASINT) uses the physical signatures of objects and events. These include infrared signatures of missile plumes and industrial activity, spectral signatures of materials (hyperspectral imaging) and radar signatures. They also include the light flash, X-rays and electromagnetic pulse of nuclear detonations. Nuclear-detonation detection sensors have flown on GPS satellites since the 1980s as part of the US Nuclear Detonation Detection System, continuing the role of the Vela satellites in verifying the Limited Test Ban Treaty.

19.4 Military Satellite Communications

Military forces need communications that are global, secure and resistant to jamming and interception. Military SATCOM is usually divided into three tiers (Table 19.4):

TABLE 19.4 Military satellite communication tiers.

Tier	Purpose	Bands	Example systems

Tier	Purpose	Bands	Example systems
Protected / strategic	Survivable, jam-resistant, low-probability-of-intercept links for nuclear command and control and critical operations	EHF (44 GHz uplink / 20 GHz downlink)	US AEHF (six satellites, 2010–2020) and planned ESS; Milstar (legacy)
Wideband	High-capacity links for headquarters, ships, aircraft and drones	X, Ka	US WGS; UK Skynet 5/6; France Syracuse 4; Italy Sicral 2; Spain SpainSat NG
Narrowband / tactical	Voice and data to small mobile terminals	UHF	US MUOS (five satellites, 2012–2016); UHF Follow-On
Commercial and hybrid	Additional capacity, resilience and Arctic coverage	Ku, Ka, L	Commercial GEO and LEO services, Starshield, Space Norway ASBM

Jam resistance is achieved mainly by spreading signals over very wide bandwidths — by direct-sequence spreading or rapid **frequency hopping** — so that a jammer must spread its power across the whole band. The protection, called **processing gain**, is $10\log(W/R)$, where W is the spread bandwidth and R the data rate (Figure 19.7). EHF systems hop over about 2 GHz, giving more than 60 dB of protection to low-rate strategic messages. Other techniques include **nulling antennas** that steer deep nulls towards jammers, on-board processing that demodulates signals and rejects interference, and narrow spot beams. **Low probability of intercept and detection (LPI/LPD)** waveforms hide transmissions below the noise.

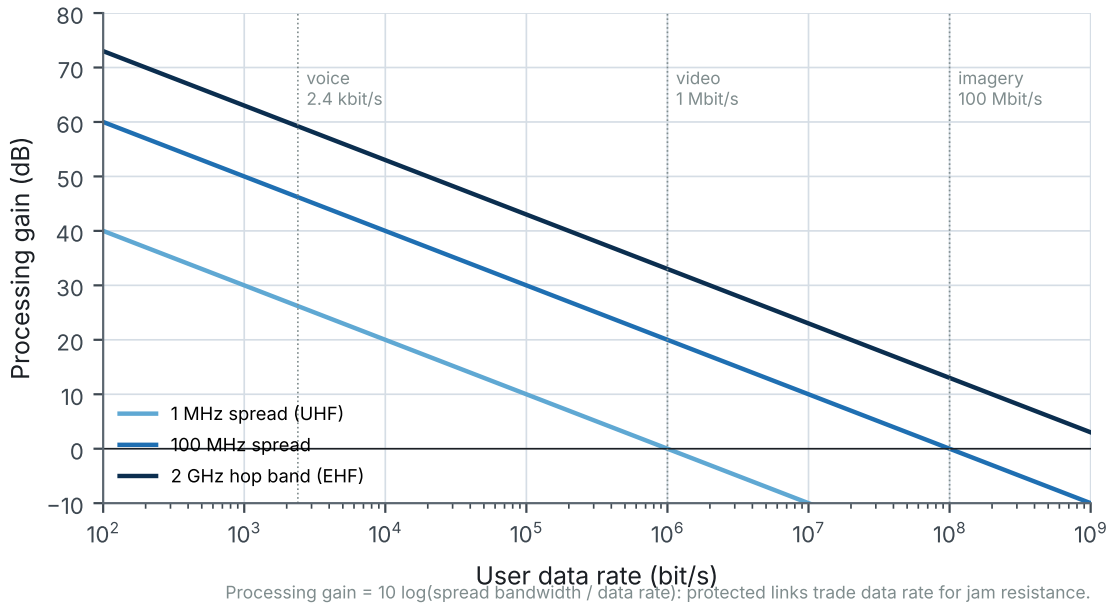


FIGURE 19.7 Processing gain of spread-spectrum military links versus user data rate. High data rates leave little margin against jamming.

The war in Ukraine demonstrated the value — and vulnerability — of commercial communications in conflict. Thousands of Starlink terminals provided resilient connectivity to Ukrainian forces and civil infrastructure. Meanwhile, on 24 February 2022, the first day of the invasion, a cyber attack on the ground network of Viasat's KA-SAT system disabled tens of thousands of terminals across Europe.

19.5 Military Weather, Launch and the Wider Enterprise

19.5.1 Military Weather

Weather affects every military operation — cloud cover over a target, winds for parachute drops, sea state for landings, and space weather for communications and GPS. The US Defense Meteorological Satellite Program (DMSP), operating since the 1960s, pioneered night-time visible imaging and microwave sounding; its last satellites are reaching the end of their lives. Its replacements include the **Weather System Follow-on – Microwave (WSF-M)**. The first WSF-M was launched in April 2024 to measure ocean-surface winds and tropical-cyclone structure. The replacements also include small electro-optical/infrared weather satellites for cloud imagery over regions where geostationary coverage is poor. Military users also rely heavily on civil systems such as Metop and JPSS (Chapter 16).

19.5.2 National Security Launch

Assured access to space is a national-security requirement in itself. The USA certifies launch providers for national-security payloads under the **National Security Space Launch (NSSL)** programme. In the 2020s, launches were shared between SpaceX (Falcon 9 and Falcon Heavy) and United Launch Alliance (Vulcan Centaur, which replaced Atlas V and Delta IV). From Phase 3 (2025 onward), they were also shared with Blue Origin's New Glenn, with a separate "lane" for new entrants on lower-risk missions. **Tactically responsive space** demonstrations, such as Victus Nox in 2023, aim to show that satellites can be replaced quickly in a crisis. In Victus Nox, a satellite was launched 27 hours after the order was given. China, Russia, India and Europe maintain their own launchers for sovereign access (Chapter 36).

19.5.3 Who Operates Military Satellites?

Figure 19.8 summarises the publicly catalogued military satellites of the main space powers by orbit. The USA operates the largest number — boosted by proliferated LEO constellations — followed by China, whose Yaogan series alone counts about 170 satellites, and Russia. Many military satellites operate in LEO for imaging and SIGINT; communications, missile-warning and some SIGINT satellites are in GEO and highly elliptical orbits; and navigation satellites are in MEO.

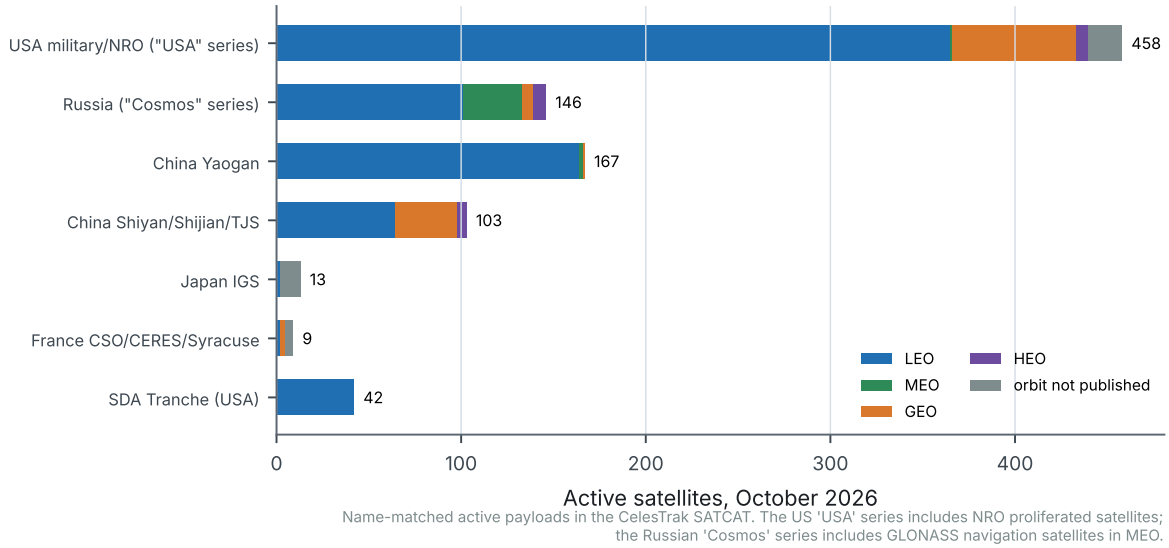


FIGURE 19.8 Active military satellites of selected series by orbit regime in October 2026 (public catalogue).

19.5.4 Allies and Commercial Partners

No country can afford all space capabilities alone. The **Combined Space Operations (CSpO)** initiative brings together the space commands of Australia, Canada, France, Germany, Italy, Japan, New Zealand, Norway, the UK and the USA to coordinate operations and share data. NATO adopted its first space policy in 2019 and established a Space Centre at its air command in Germany. Commercial companies increasingly act as partners: providing imagery, RF geolocation, communications, launch and space-tracking data, and in some cases building dedicated government constellations. This raises new questions — such as whether commercial satellites supporting military operations become legitimate targets, and how governments protect companies that operate in conflict zones.

19.6 From Exquisite to Proliferated Architectures

For most of the space age, national-security satellites were few, large, expensive and long-lived — "exquisite" systems that took a decade to develop and cost billions of dollars each. Such systems are extremely capable but also tempting targets: losing one can remove a capability for years. Since about 2019, the USA and other countries have shifted towards **resilience** through:

- **Proliferation**: many cheaper satellites, so that losing a few does not matter;
- **Disaggregation**: separating functions (for example strategic and tactical missile warning) onto different satellites;
- **Diversification**: using several orbits, frequencies and providers, including commercial ones;
- **Rapid replenishment**: replacing satellites every few years in "tranches" that incorporate new technology.

The US **Space Development Agency (SDA)**, founded in 2019, is building the **Proliferated Warfighter Space Architecture (PWSA)** in low Earth orbit at about 1 000 km (Figure 19.9). Its **Transport Layer** forms a mesh network of optical inter-satellite links and radio links (including the Link 16 tactical data link) to deliver targeting data directly to forces. Its **Tracking Layer** carries infrared sensors to detect and track missiles, including hypersonic glide vehicles

(Chapter 20). About two dozen Tranche 0 demonstration satellites launched in 2023, and the first operational Tranche 1 satellites launched in September 2025. Tranche 1 comprises 154 satellites (126 transport and 28 tracking), with further tranches planned every two years. Separately, the NRO has been deploying a proliferated constellation of imaging satellites, built with SpaceX and Northrop Grumman, since 2024.

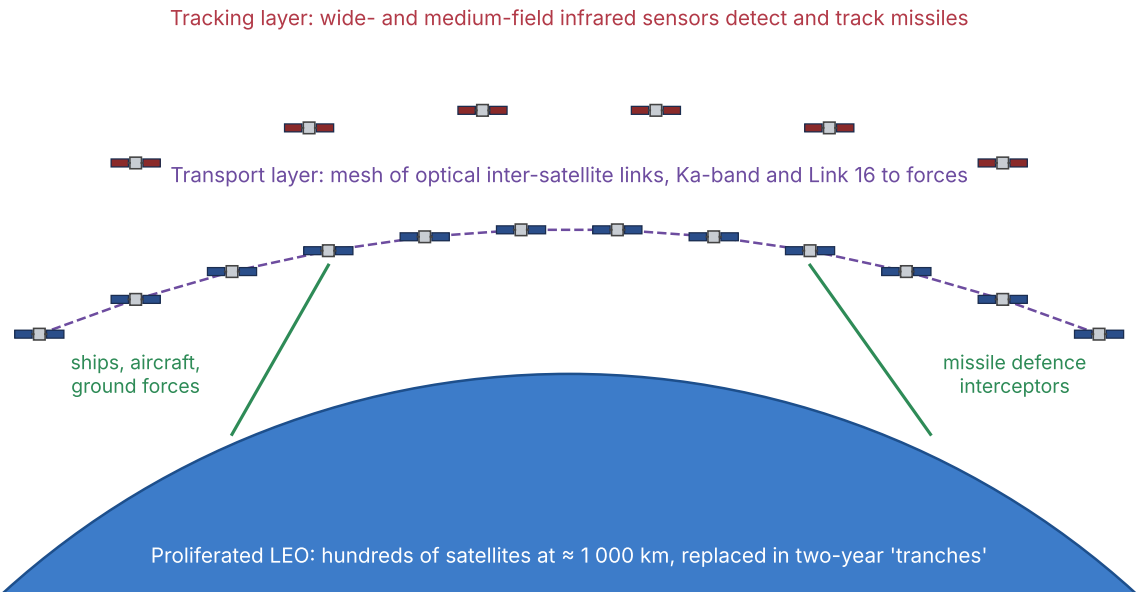


FIGURE 19.9 The Proliferated Warfighter Space Architecture: a transport layer of laser-linked satellites delivers data from a tracking layer of infrared sensors to forces and missile-defence systems.

In 2025 the US government launched **Golden Dome**, an ambitious homeland missile-defence initiative that includes space-based sensors and, controversially, **space-based interceptors**. In April 2026 the Space Force awarded prototype agreements worth up to $3.2 billion to twelve companies to demonstrate space-based interceptors by 2028; independent analysts have questioned the technical feasibility and cost of the overall programme, which official estimates put at about $175–185 billion.

Other powers are following similar paths: China has launched large numbers of Yaogan and other military satellites and is building proliferated LEO constellations, and Europe's IRIS^2 constellation will include secure governmental communications.

19.6.1 Spaceplanes and Experimental Platforms

The US Space Force's **X-37B** Orbital Test Vehicle, a small reusable uncrewed spaceplane built by Boeing (Chapter 10), has flown eight missions since 2010, some lasting more than two years. Its seventh mission (2023–2025) demonstrated aerobraking from a highly elliptical orbit, and its eighth was launched in August 2025. China has flown a comparable reusable experimental spacecraft several times since 2020. Such platforms test new technologies and allow sensors to be returned to Earth for inspection.

19.7 Counterspace: A Contested Domain

19.7.1 Categories of Threat

As military dependence on space grew, so did interest in denying an adversary's use of it. Public assessments by organisations such as the Secure World Foundation and the Center for Strategic and International Studies group counterspace capabilities into four categories (Figure 19.10):

- **Kinetic physical**: direct-ascent anti-satellite (DA-ASAT) missiles that destroy a satellite by collision; co-orbital weapons that approach and attack a target; attacks on ground stations.
- **Non-kinetic physical**: lasers that dazzle or damage optical sensors, high-power microwaves and — most dangerously — nuclear detonations in space, whose radiation would damage large numbers of satellites indiscriminately.
- **Electronic**: jamming of uplinks and downlinks and spoofing of GNSS and other signals — the most widely used and most easily reversible form of counterspace.
- **Cyber**: attacks on the data, ground systems or satellites themselves.

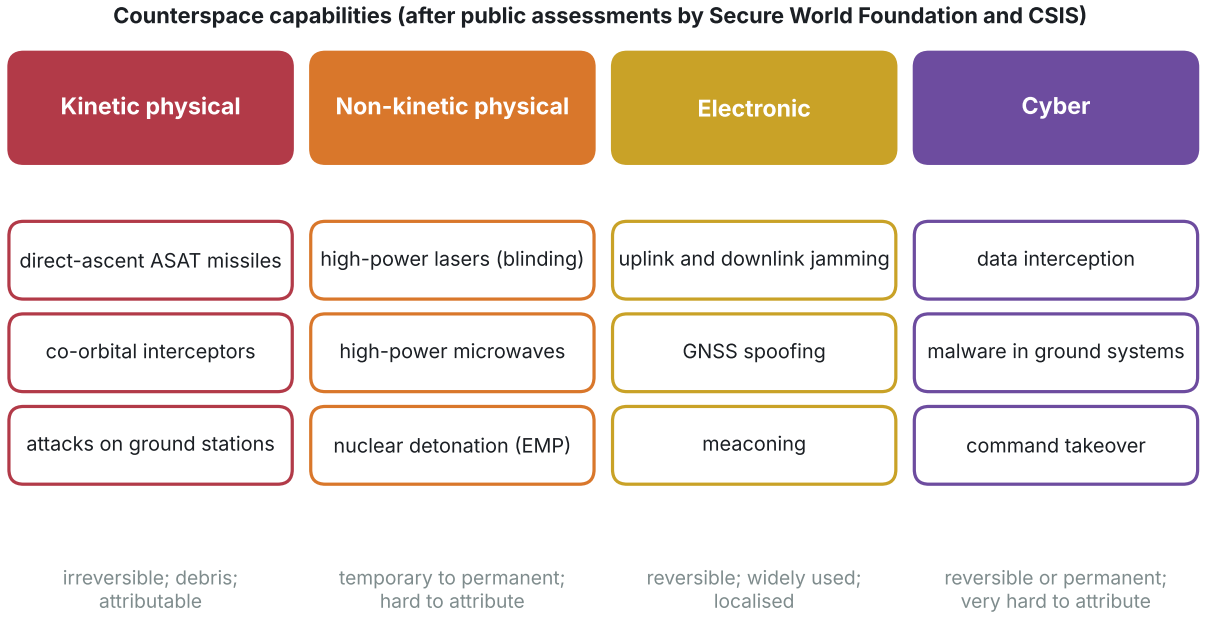


FIGURE 19.10 Categories of counterspace capability.

19.7.2 Anti-Satellite Tests and Debris

Five destructive anti-satellite tests have created long-lived debris (Figure 19.11; Table 19.5). The worst was China's January 2007 destruction of its Fengyun-1C weather satellite at about 865 km. It produced more than 3 500 catalogued fragments — the largest debris-generating event in history. More than 2 300 were still in orbit in October 2026, threatening satellites in sun-synchronous orbits for decades. Russia's November 2021 test against the defunct Cosmos 1408 at about 480 km produced about 1 800 catalogued fragments and forced the ISS crew to shelter in their spacecraft; most have since re-entered. Tests at lower altitudes included the US destruction of the failing USA-193 satellite in 2008 at about 247 km and India's "Mission Shakti" in 2019 at about 283 km. These created debris that decayed within months to a few years.

TABLE 19.5 Destructive direct-ascent anti-satellite tests (debris counts from the CelesTrak SATCAT).

Year	Country	Target	Intercept altitude	Catalogued debris	Still in orbit (Oct 2026)
1985	USA	Solwind P78-1 (air-launched ASM-135)	≈ 525 km	286	0
2007	China	Fengyun-1C (SC-19 missile)	≈ 865 km	3 534	2 314
2008	USA	USA-193 (SM-3 from a ship)	≈ 247 km	174	0
2019	India	Microsat-R (PDV Mk-II)	≈ 283 km	129	0
2021	Russia	Cosmos 1408 (Nudol)	≈ 480 km	1 806	4

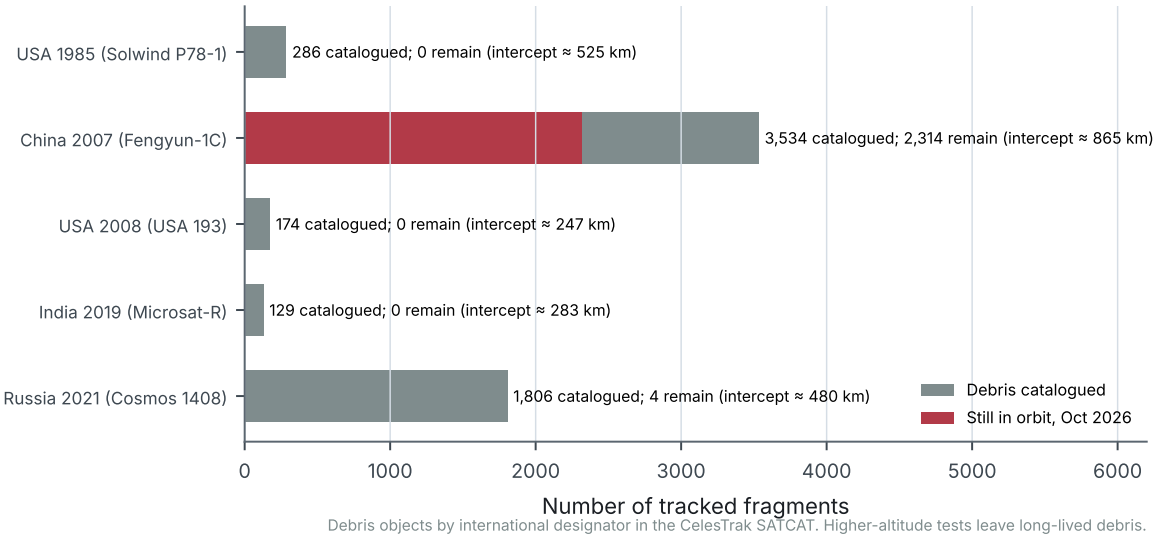
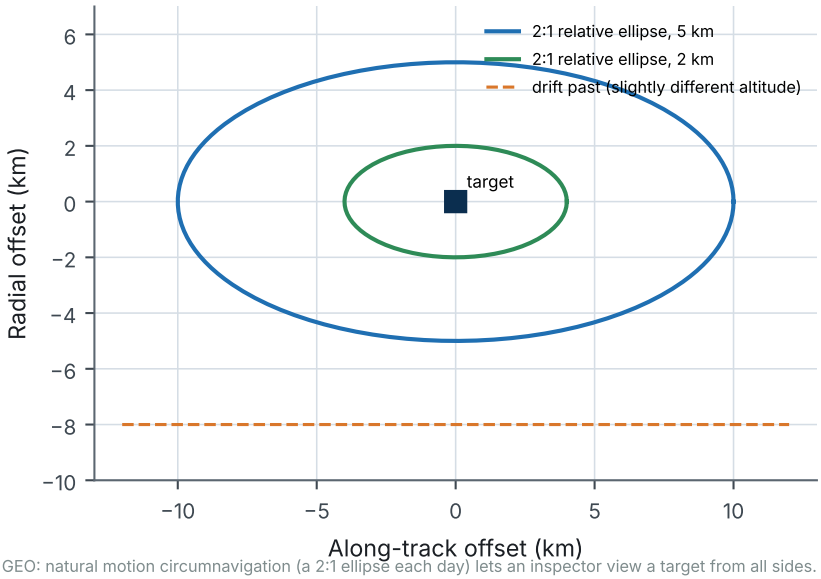


FIGURE 19.11 Tracked debris from destructive anti-satellite tests. Debris from the high-altitude 2007 test remains in orbit; debris from lower-altitude tests has re-entered.

In April 2022 the USA announced a unilateral commitment not to conduct destructive direct-ascent ASAT missile tests. In December 2022 the UN General Assembly adopted a resolution calling on all states to make the same commitment. Dozens of countries have since joined the pledge.

19.7.3 Rendezvous and Proximity Operations

Satellites that can approach other satellites — for inspection, servicing or, potentially, attack — are a growing concern because the same technology serves peaceful and hostile purposes. The USA operates the Geosynchronous Space Situational Awareness Program (GSSAP) satellites to inspect objects in GEO. Russian satellites such as Cosmos 2542 and Luch/Olymp have caused public concern. Cosmos 2542 released a sub-satellite that manoeuvred near a US reconnaissance satellite in 2020, and Luch/Olymp has parked near Western communication satellites. China's Shijian-21 towed a dead BeiDou satellite to a graveyard orbit in 2022, demonstrating capture technology. Orbital mechanics shapes these operations: an inspector at a slightly different orbit can perform a natural circumnavigation of its target once per orbit without using propellant (Figure 19.12).



Clohessy–Wiltshire relative motion in GEO: natural motion circumnavigation (a 2:1 ellipse each day) lets an inspector view a target from all sides.

FIGURE 19.12 Relative motion of an inspector satellite around a target in geostationary orbit (Clohessy–Wiltshire equations). A suitably offset orbit produces a daily 2:1 ellipse around the target.

19.7.4 Jamming, Spoofing and Cyber Attacks

Electronic and cyber attacks are now routine in conflicts. GNSS jamming and spoofing affect aviation and shipping across large regions (Chapter 18), and satellite television and communications are jammed. The 2022 KA-SAT attack showed how a cyber attack on a satellite operator's ground network can have international effects. Because these attacks are reversible, hard to attribute and often below the threshold of armed conflict, they are difficult to deter.

19.7.5 Nuclear Weapons in Space

The 1962 US Starfish Prime test, a 1.4-megaton nuclear explosion at 400 km, created artificial radiation belts that damaged or destroyed about a third of the satellites then in orbit (Chapter 22). The Outer Space Treaty (1967) prohibits placing nuclear weapons or other weapons of mass destruction in orbit. In 2024 US officials publicly expressed concern that Russia was developing a nuclear anti-satellite capability; a UN Security Council resolution reaffirming the prohibition was vetoed by Russia in April 2024.

19.8 Law, Norms and Stability

International law applies in space. The **Outer Space Treaty** declares space the province of all humankind and prohibits national appropriation and weapons of mass destruction in orbit. It also requires states to authorise and supervise their nationals' space activities. The UN Charter, including the right of self-defence, and the law of armed conflict also apply. However, there is no treaty banning conventional weapons in space or ground-based anti-satellite weapons. Diplomatic efforts include:

- Russian–Chinese proposals for a treaty on the Prevention of the Placement of Weapons in Outer Space (PPWT), criticised by Western states for lacking verification and ignoring ground-based ASATs;
- the UK-led UN Open-Ended Working Group on **reducing space threats through norms, rules and principles of responsible behaviour** (2022–23);

- the destructive DA-ASAT test moratorium;
- expert efforts to clarify how existing law applies, such as the Woomera Manual and the MILAMOS project.

Transparency measures include announcing manoeuvres, sharing space-tracking data, and establishing communication channels between space commands. They aim to reduce the risk of misunderstanding in a crowded orbital environment, where an accidental collision could be mistaken for an attack.

EXAMPLE: GEOLOCATING A SHIP RADAR WITH TWO SATELLITES

Two SIGINT satellites fly 80 km apart at 600 km altitude. A ship's navigation radar transmits pulses at 9.4 GHz.

1. The satellites time-tag each received pulse with GNSS-disciplined clocks. A time difference of 100 µs means the ship is about 30 km farther from one satellite than the other; the ship lies on a hyperbola-like curve on the sea surface (Figure 19.5, left).
2. A timing precision of 50 ns corresponds to a range-difference precision of 15 m; geometry stretches the position error along the curve, typically to a few hundred metres.
3. The satellites also measure the Doppler shift of the pulses: their slightly different velocity vectors give a frequency difference of a few kilohertz that places the ship on a second family of curves (Figure 19.5, right).
4. The intersection gives the ship's position. Repeating the measurement on successive pulses, or on the next pass, gives its course and speed — even if it has switched off its AIS transponder.

AI IN NATIONAL-SECURITY SPACE

AI is used to detect objects and changes in imagery, classify radio emitters from their signal characteristics, fuse data from many sensors, detect anomalous satellite behaviour, plan collections across large constellations and support battle management. These uses raise questions about reliability, adversarial manipulation and human control, discussed in Chapters 50 and 53.

KEY TAKEAWAYS

- National-security space includes ISR, missile warning, PNT, SATCOM, weather, space domain awareness, space control and launch.
- Reconnaissance satellites date from 1960 (Corona film return, GRAB SIGINT); imagery is now digital and complemented by large commercial constellations.
- SIGINT satellites intercept radar (ELINT), communications (COMINT) and telemetry (FISINT); emitters are located by angle of arrival, TDOA and FDOA, often with satellite clusters. Commercial RF geolocation now serves maritime security.
- Military SATCOM uses wide spreading and frequency hopping for jam resistance; commercial services have become important but are also targets.
- Architectures are shifting from a few exquisite satellites to proliferated, frequently replenished constellations (SDA PWSA) for resilience.
- Five destructive ASAT tests have created thousands of debris fragments; the 2007 Chinese test left debris that will persist for decades. Jamming, spoofing and cyber attacks are now routine in conflicts.
- International law applies in space, but there is no ban on conventional space weapons; norms such as the DA-ASAT test moratorium aim to improve stability.

REVIEW QUESTIONS

1. List the mission areas of national-security space and give one example system for each.
2. Why was Corona so important, and how did it work?
3. Distinguish ELINT, COMINT and FISINT.
4. Two satellites measure a time difference of arrival of 2.0 μs for the same radar pulse. What path-length difference does this represent, and what locus on the Earth's surface does it define?
5. How does FDOA geolocation work, and why do commercial RF-geolocation companies fly satellites in clusters?
6. Explain why national-security architectures are moving from a few exquisite satellites to proliferated constellations.
7. Why are destructive anti-satellite tests especially harmful to all space users? Use the 2007 test as an example.
8. What legal principles apply to military activities in space, and what is the purpose of the moratorium on destructive direct-ascent ASAT tests?

20 CHAPTER Missile Early Warning Satellites



Image: NASA/JSC

The fiery exhaust of a rocket is one of the brightest infrared sources on Earth. Since 1970, satellites watching for that signature have given the first warning of ballistic-missile launches anywhere in the world, underpinning nuclear deterrence and, increasingly, missile defence. This chapter explains, from public sources, how early-warning satellites work: the infrared physics of rocket plumes and the atmosphere, scanning and staring sensors, orbits that cover the globe and the poles, the processing that separates launches from sunlight and clouds, and the timelines of ballistic-missile flight. It traces the history from MIDAS and the Defense Support Program to SBIRS and Next-Generation OPIR, describes Russian, Chinese and European systems, and explains why hypersonic glide vehicles are driving a shift to proliferated low-orbit tracking constellations.

IN THIS CHAPTER

20.1 Why Warning from Space?
20.2 History
20.3 The Physics of Seeing Rockets
20.4 Sensors and Satellites
20.5 Missile Flight and Warning Timelines
20.6 Hypersonic Glide Vehicles and the Tracking Layer
20.7 Systems in Operation
20.8 Beyond Missiles: Other Infrared Events
20.9 Stability and Arms Control

20.1 Why Warning from Space?

A ballistic missile launched from one continent can reach another in about 30 minutes; a missile launched from a submarine near the coast can arrive in 10–15 minutes. Ground-based radars, limited by the curvature of the Earth, can see such missiles only after they rise above the horizon

— typically several minutes into flight and thousands of kilometres from the launch point. A satellite high above the Earth, by contrast, can see a launch almost anywhere within seconds, because the rocket's exhaust plume is an intense source of infrared radiation.

Early-warning satellites therefore serve four purposes:

1. **Strategic warning**: detecting missile attacks on a nation's territory and confirming that an attack is real, a cornerstone of nuclear deterrence and crisis stability.
2. **Missile defence**: providing the earliest possible track of a missile to cue radars and interceptors, and increasingly providing tracks precise enough to guide interceptors directly.
3. **Battlespace awareness**: detecting and characterising other infrared events — tactical missile launches, explosions, aircraft afterburners, large fires.
4. **Technical intelligence**: measuring the performance of foreign missiles during tests.

The same satellites, together with sensors on GPS satellites, also detect nuclear detonations.



FIGURE 20.1 The Defense Support Program satellite DSP Flight 16 on its Inertial Upper Stage in the payload bay of Space Shuttle Atlantis before deployment in November 1991. DSP satellites provided US missile warning for almost five decades. Image: NASA/JSC.

20.2 History

The US **MIDAS** (Missile Defense Alarm System) programme launched twelve experimental satellites between 1960 and 1966 (Figure 20.2). Many failed, and early sensors were overwhelmed by false alarms from sunlight reflected by clouds, but MIDAS proved that rocket launches could be detected from orbit. Its successor, the **Defense Support Program (DSP)**, launched its first satellite in 1970; 23 DSP satellites were launched until 2007, and some operated for more than 20 years. DSP gained public fame in the 1991 Gulf War, when it detected Iraqi Scud launches and warned Coalition forces and Israel.

The Soviet Union developed the **Oko** system from the 1970s, using satellites in highly elliptical Molniya orbits. These viewed US missile fields obliquely, against the dark background of space near the Earth's limb, to avoid confusion with sunlit clouds. On 26 September 1983 an Oko satellite falsely reported the launch of several US missiles — sunlight reflected off high-altitude clouds had triggered the alarm. The duty officer, Lieutenant Colonel Stanislav Petrov, judged it to be a false alarm and did not pass it up the chain of command. This episode is often cited as an example of the dangers of automated warning systems and the importance of human judgement.

The US **Space-Based Infrared System (SBIRS)** replaced DSP: sensors hosted on classified satellites in highly elliptical orbits from 2006–2008, and six SBIRS GEO satellites launched between 2011 and 2022. Russia replaced Oko with the **EKS (Tundra/Kupol)** system from 2015, and China, France and others have developed their own capabilities.

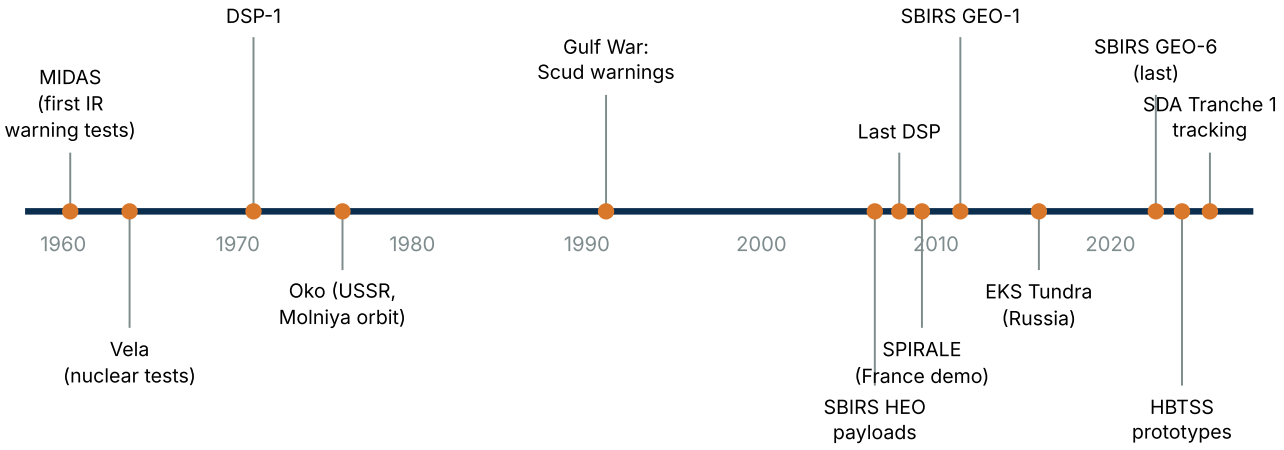
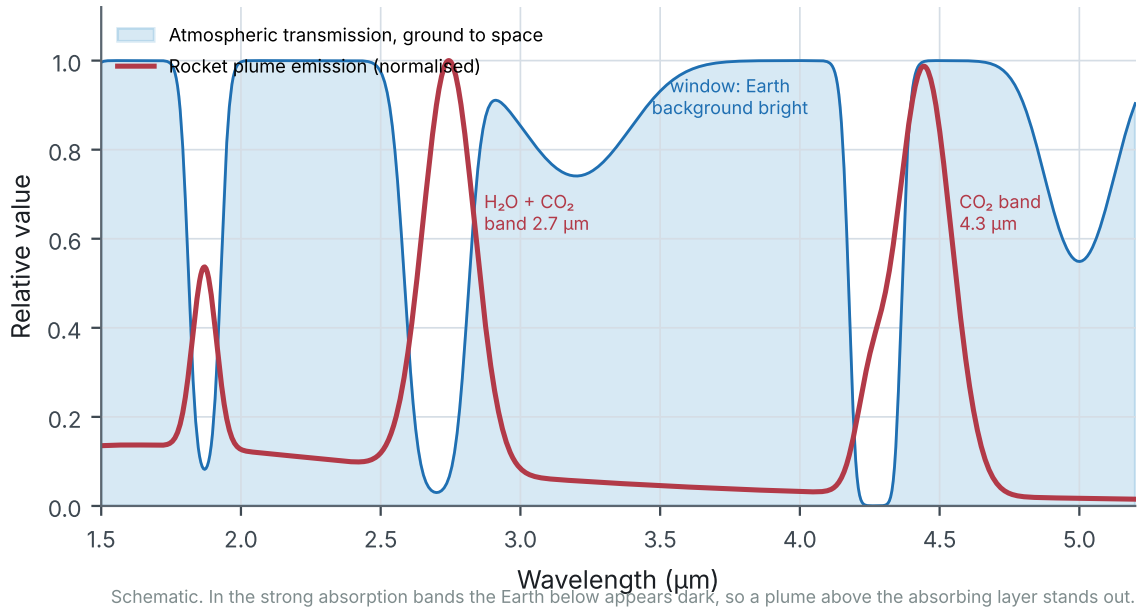


FIGURE 20.2 Milestones in space-based missile warning.

20.3 The Physics of Seeing Rockets

20.3.1 Plume Emission

A rocket plume consists of hot combustion gases — mainly water vapour, carbon dioxide, carbon monoxide and, for solid-propellant motors, aluminium oxide particles — at temperatures of 1 500–3 000 K. Hot gases emit strongly in the vibrational bands of their molecules, especially near **2.7 µm** (H_2O and CO_2) and **4.3 µm** (CO_2), while hot particles emit a continuum like a grey body (Figure 20.3). Where the plume mixes with the atmosphere, unburned fuel continues to burn ("afterburning"), increasing its size and brightness. A large booster plume radiates megawatts per steradian in the infrared.



Schematic. In the strong absorption bands the Earth below appears dark, so a plume above the absorbing layer stands out.

FIGURE 20.3 Schematic emission of a rocket plume compared with the transmission of the atmosphere. The strongest plume emission falls in molecular bands where the lower atmosphere is opaque.

20.3.2 The Atmosphere as a Filter

The atmosphere is opaque in the same bands where the plume is brightest, because the same molecules — water vapour and carbon dioxide — absorb there. This is the key to early-warning sensor design. Suppose a sensor looks at the Earth in a strong absorption band such as 2.7 µm. The radiation from the ground, clouds and sunlit surfaces is then absorbed before it reaches space, so the Earth appears dark and uniform. A rocket plume is invisible while it is low in the atmosphere. The absorbing layer extends a few kilometres up for the water-vapour band, and higher in the more strongly absorbing parts of the CO_2 band. Once the plume climbs above this layer, it suddenly appears as a bright point against a dark background (Figure 20.4). This **"above-the-horizon of absorption"** approach suppresses clutter from clouds and sun glints; sensors working in more transparent bands see launches earlier but must cope with a bright, structured background.

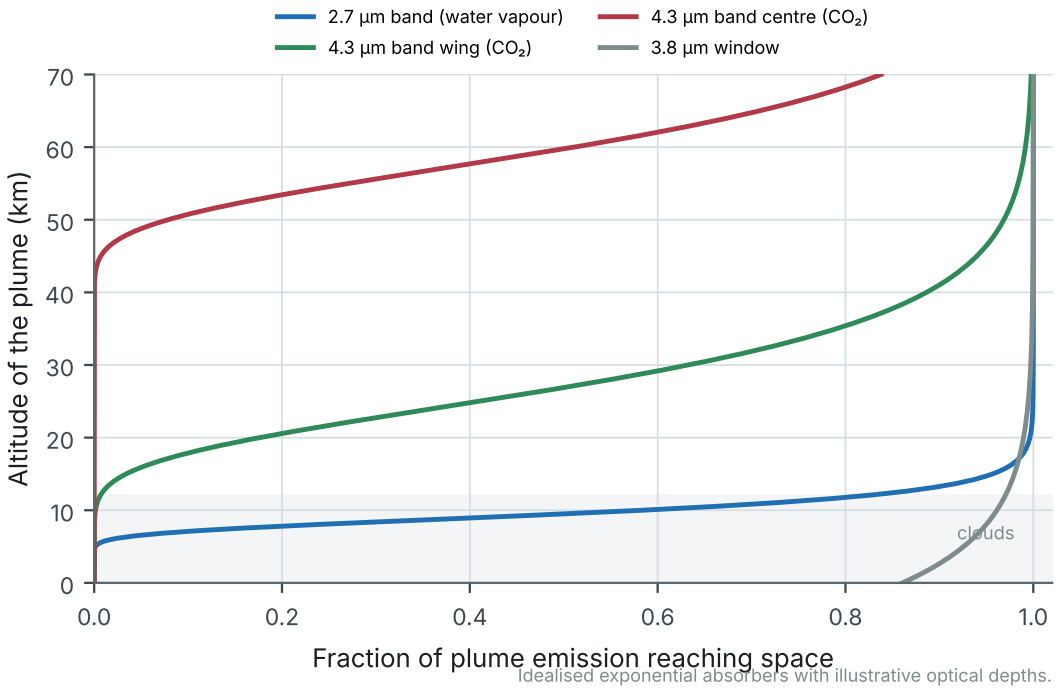


FIGURE 20.4 Fraction of plume radiation escaping to space as a function of the plume's altitude in different spectral bands. In strongly absorbing bands, a plume becomes visible only after it climbs above the absorbing layer, but the background below is suppressed.

20.3.3 How Far Can Sensors See?

The irradiance at the sensor falls with the square of the distance (Figure 20.5). A bright ICBM booster seen from geostationary orbit, 36 000–40 000 km away, delivers an irradiance far above the sensitivity of modern infrared sensors. Dimmer targets include short-range missiles, the hot bodies of re-entry vehicles after burnout, and hypersonic glide vehicles. These are much harder to see from GEO, and they favour sensors in lower orbits, which are 30–40 times closer.

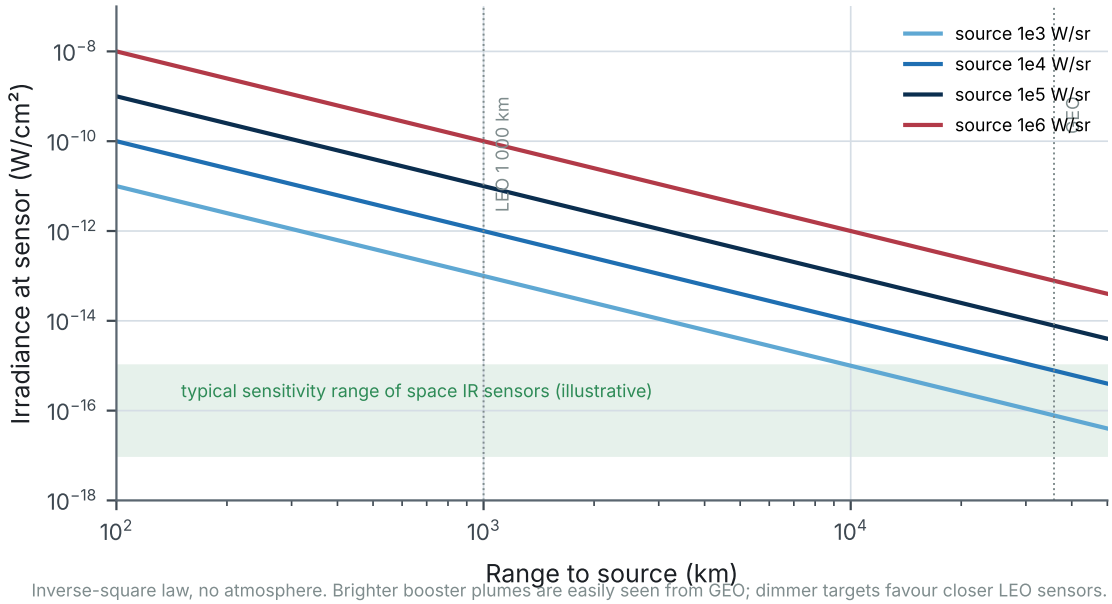


FIGURE 20.5 Irradiance from infrared sources of different intensity as a function of range, compared with an illustrative sensitivity range for space infrared sensors.

20.4 Sensors and Satellites

20.4.1 Scanning and Staring

Two basic sensor architectures are used (Figure 20.6):

- **Scanning sensors**, as on DSP, sweep a linear array of detectors across the scene. DSP satellites rotated about six times per minute. A telescope tilted slightly off the spin axis swept a line of several thousand lead-sulphide (and later also mercury-cadmium-telluride) detectors around the Earth's disc. Each point was revisited about every 10 seconds. Scanning is efficient with small detector arrays but cannot follow a fast event continuously.
- **Staring sensors**, made possible by large two-dimensional infrared focal-plane arrays, watch an area continuously, measuring the brightness of every pixel many times per second. They detect dimmer and shorter events and measure the time history of a plume precisely. SBIRS GEO combined a **scanning sensor** covering the whole Earth with a **staring sensor** that could be pointed at areas of interest; Next-Gen OPIR uses large staring arrays.

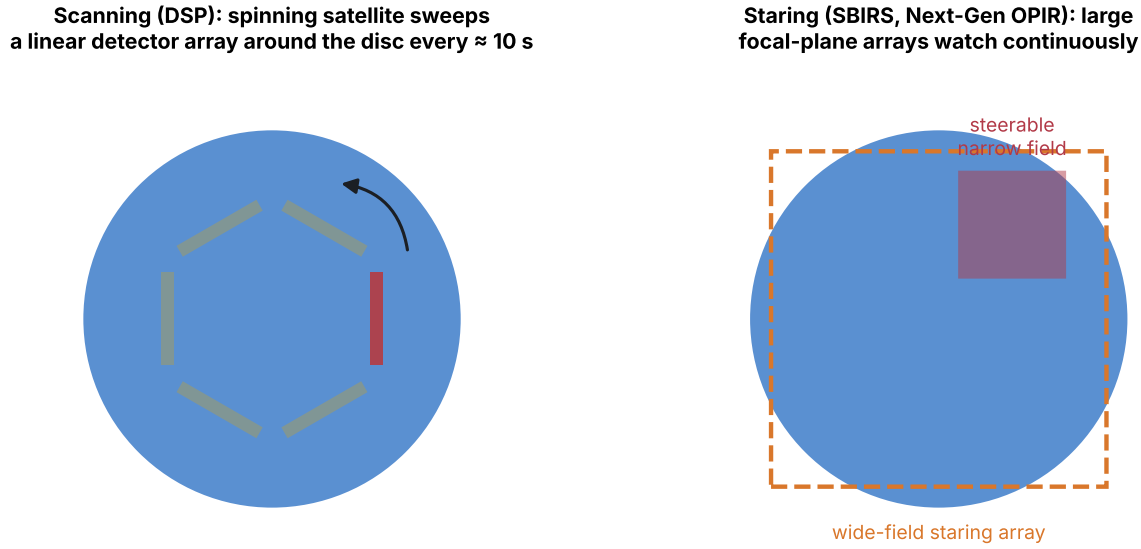


FIGURE 20.6 Scanning (DSP) and staring (SBIRS, Next-Gen OPIR) sensor architectures.



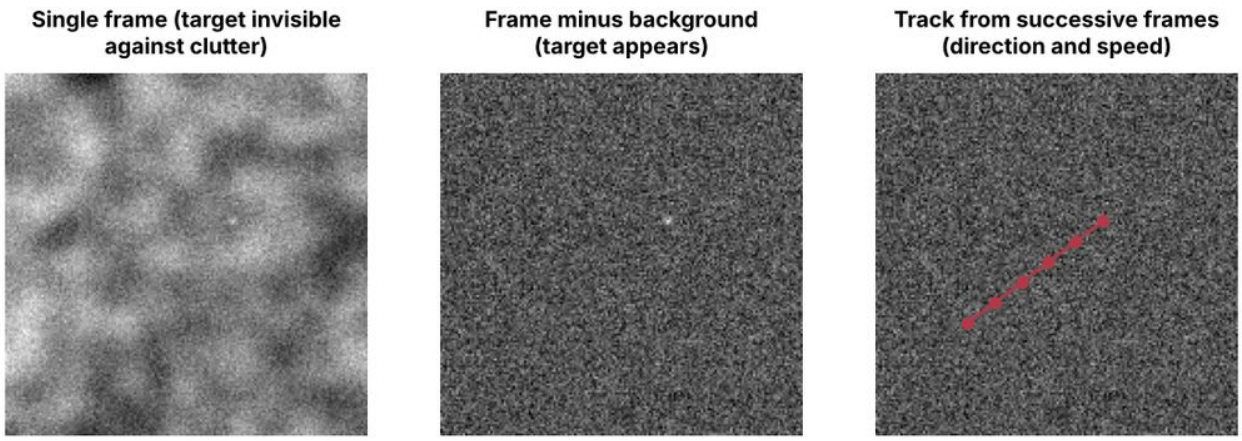
FIGURE 20.7 A DSP satellite being prepared at Cape Canaveral in 1991 for launch on the Space Shuttle. The large telescope tube housing the infrared sensor projects from the spacecraft; the satellite spun around its axis to scan the Earth. Image: NASA/JSC.

20.4.2 Detection and Clutter Rejection

Turning detector signals into reliable warnings is a demanding signal-processing problem. The background — sunlight reflected by clouds and water (sun glint), hot surfaces, fires, volcanoes, lightning and even high-altitude aircraft — produces vastly more "events" than missile launches. On-board and ground processing:

- estimate and subtract the slowly varying background in each pixel;
- detect pixels whose brightness rises above a threshold (constant-false-alarm-rate detection);
- associate detections in successive frames into **tracks** with consistent motion (Figure 20.8);
- compare the track's intensity history and multi-band "colour" with the signatures expected from missiles;
- fuse data from two or more satellites to compute the three-dimensional trajectory by stereo triangulation, estimate the launch point and predict the impact area.

Within a minute or two of launch, a missile-warning centre (for the USA, at Buckley Space Force Base in Colorado) can report the launch, the missile type and its predicted impact region.



Simulated infrared frames: a faint moving source is detected by background subtraction and temporal tracking.

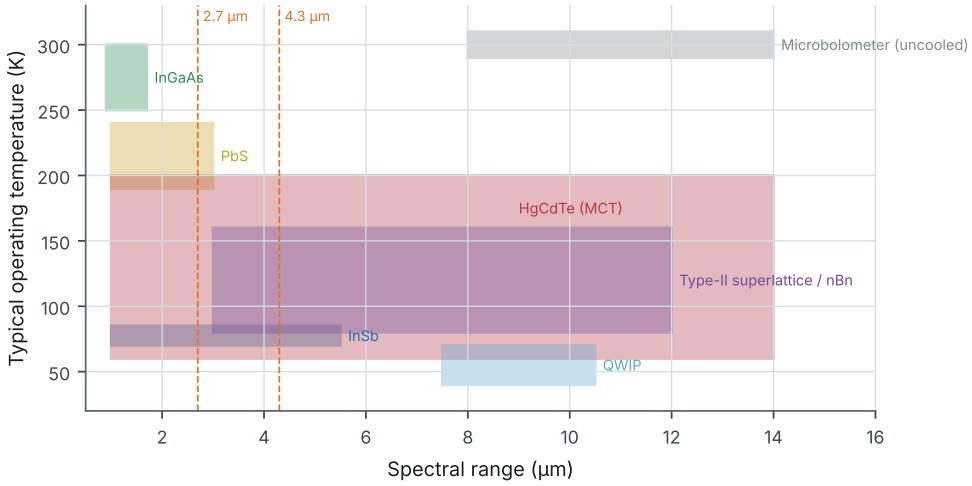
FIGURE 20.8 Simulated detection of a faint moving infrared source: invisible in a single frame against structured clutter, it emerges after background subtraction and is confirmed by tracking it over successive frames.

20.4.3 Infrared Detectors

The heart of every early-warning sensor is its infrared **focal-plane array**: a two-dimensional array of photodetectors bonded to a silicon read-out integrated circuit that amplifies and digitises each pixel's signal. Detector materials are chosen for the spectral band (Figure 20.9):

- **Lead sulphide (PbS)**, used on early DSP sensors, responds to the 2.7 µm band at moderate cooling.
- **Indium antimonide (InSb)** covers 1–5.5 µm and must be cooled to about 77 K.
- **Mercury cadmium telluride (HgCdTe, "MCT")** can be tuned by its composition to cut off anywhere from about 1.7 to 14 µm; it is the workhorse for space infrared sensors, from SBIRS to astronomical telescopes.
- **Type-II strained-layer superlattices** and **nBn barrier detectors** are newer III–V semiconductor technologies that can operate at higher temperatures with lower dark current, easing cooling requirements.

Cooling is provided by passive radiators or mechanical cryocoolers (Chapter 5). Arrays with millions of pixels, read out hundreds of times per second, are now available — enabling wide-field staring sensors that were impossible in the DSP era. Radiation hardening is essential: energetic particles striking the array create false "spikes" that processing must reject, especially in highly elliptical orbits that pass through the radiation belts.



Approximate ranges. Longer-wavelength photon detectors must be cooled to suppress dark current; missile-warning sensors use SWIR and MWIR bands.

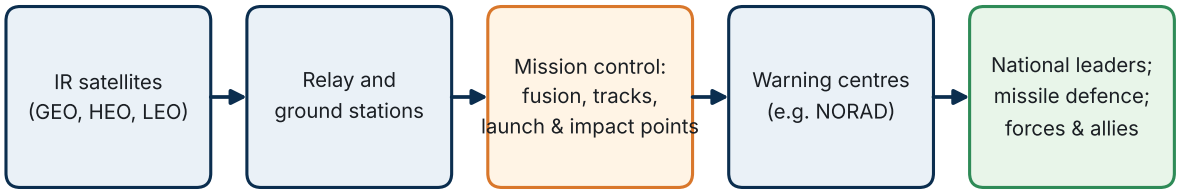
FIGURE 20.9 Approximate spectral ranges and operating temperatures of infrared detector technologies.

TABLE 20.1 Infrared detector materials.

Material	Spectral range	Operating temperature	Use in missile warning
PbS	1–3 µm	≈ 190–240 K	Early DSP sensors (2.7 µm band)
InSb	1–5.5 µm	≈ 77 K	MWIR sensors
HgCdTe	1–14 µm (tunable)	≈ 60–200 K	SBIRS, Next-Gen OPIR, LEO tracking sensors
Type-II superlattice / nBn	3–12 µm	≈ 80–160 K	Newer MWIR and LWIR sensors; higher operating temperature
QWIP	7–11 µm	≈ 40–70 K	LWIR (e.g. Landsat TIRS) — not typical for warning

20.4.4 From Detection to Decision

Missile-warning data flow from the satellites through relay and ground stations to mission-control centres. There, data from several satellites are fused into tracks with estimated launch points and impact areas. The data then flow to warning and command centres (in North America, the North American Aerospace Defense Command, NORAD) and to missile-defence systems and allied forces (Figure 20.10). The whole chain must be fast, highly reliable and protected against cyber attack and jamming. It is tested continuously with real launches, including the world's routine space launches, which look exactly like missile launches during boost.

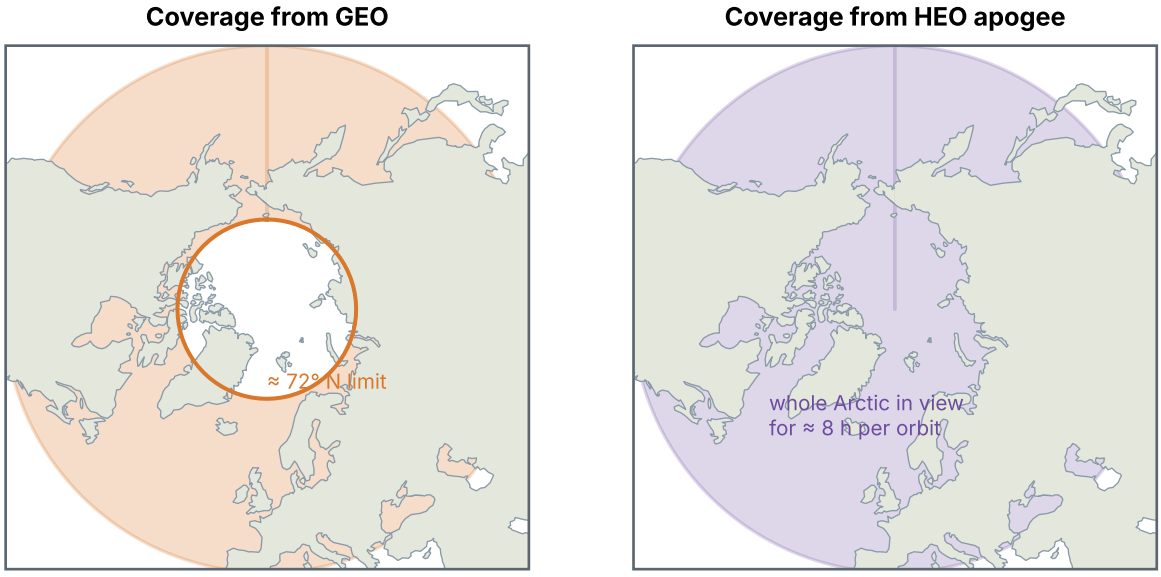


Target timelines: detection to report in ≈ 1–3 minutes; continuous updates through the flight

FIGURE 20.10 The missile-warning data chain from satellites to decision-makers.

20.4.5 Orbits

Geostationary satellites are the backbone of missile warning: three or four can watch almost the whole Earth between about 70° N and 70° S continuously. They see poorly at very high latitudes, however — exactly where the shortest routes for ICBMs between North America and Eurasia and many submarine patrol areas lie. Sensors in **highly elliptical orbits** (Molniya- or Tundra-type), which spend many hours near apogee over the Arctic, fill this gap (Figure 20.11). Russia's Oko and EKS systems rely mainly on such orbits; the USA has used sensors hosted on satellites in highly elliptical orbits since 2006 and plans dedicated Next-Gen OPIR Polar satellites.



Schematic: GEO sensors view poorly beyond ≈ 70–75° latitude; Molniya/Tundra-type orbits fill the polar gap.

FIGURE 20.11 Coverage of the Arctic from geostationary orbit and from the apogee of a highly elliptical orbit (schematic).

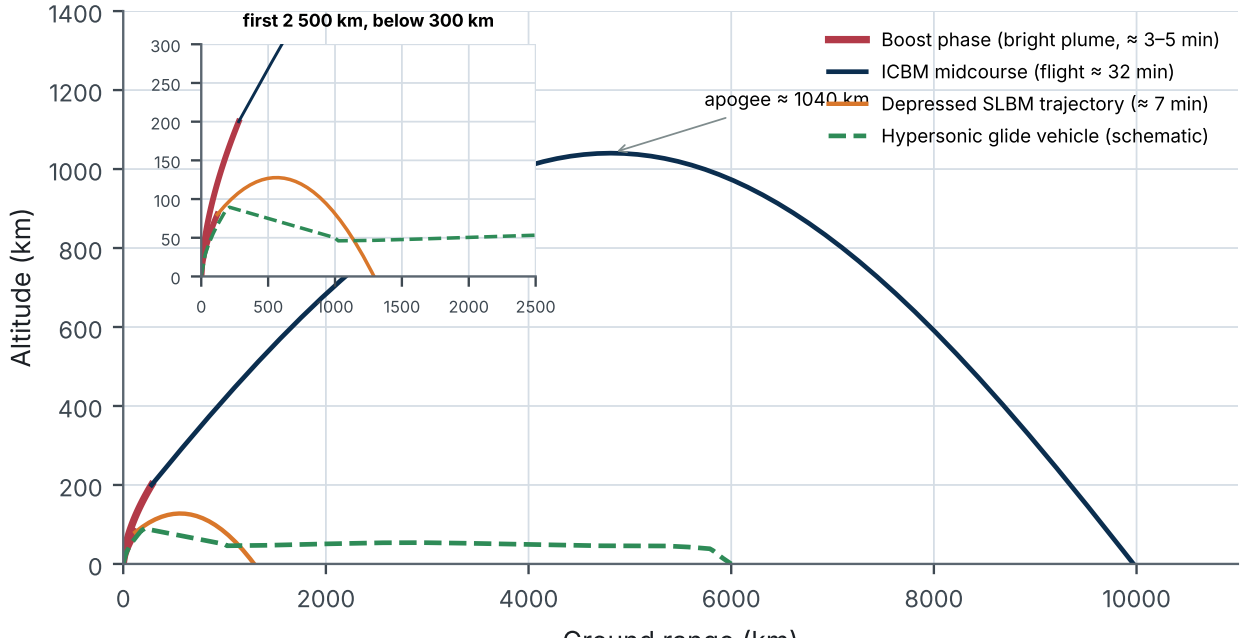
20.5 Missile Flight and Warning Timelines

20.5.1 Phases of Ballistic-Missile Flight

A ballistic missile's flight has three phases (Figure 20.12):

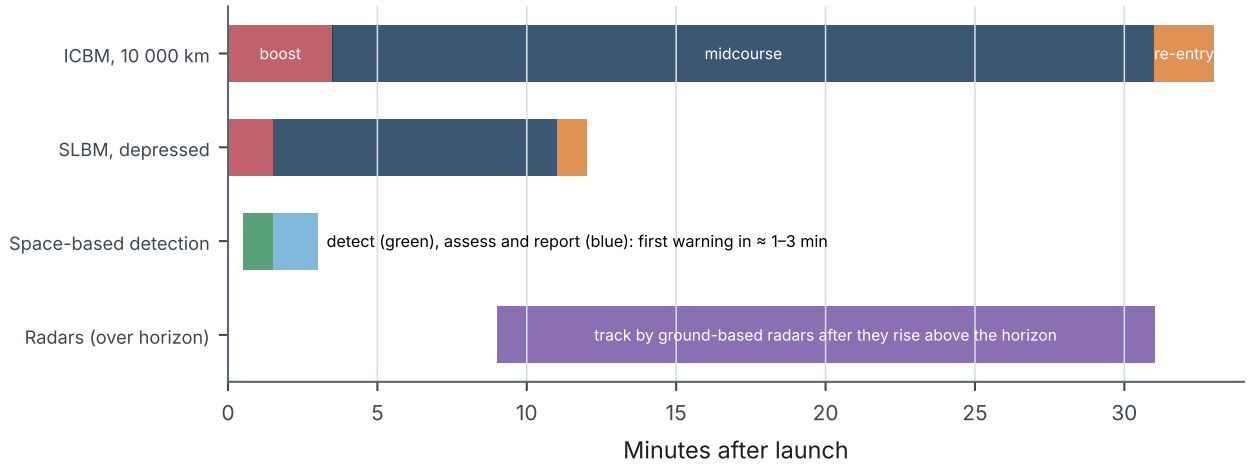
- **Boost**: the rocket engines burn for about 1–5 minutes, accelerating the payload to about 7 km/s for an intercontinental missile. This is when the plume is brightest and space-based infrared sensors see it best.
- **Midcourse**: after burnout, the warhead and any decoys coast above the atmosphere on an elliptical arc. For a minimum-energy ICBM trajectory over 10 000 km, they reach an apogee of about 1 000–1 300 km and take 20–25 minutes. Without a plume, the objects are cold and dim; they are tracked mainly by radars and by sensitive sensors observing their thermal emission against cold space.
- **Terminal**: re-entry vehicles plunge back through the atmosphere in the final minute or so.

Submarine-launched missiles can fly **depressed trajectories** — lower and faster — reducing warning time to about 10 minutes. These timelines (Figure 20.13) leave national leaders only minutes to assess an attack, which is why fast, reliable warning from space is so important for crisis stability.



Two-body trajectories after burnout (ICBM: 7.0 km/s at 17°; SLBM: 4.0 km/s at 12°). Glide vehicles fly low and manoeuvre, so they are harder to see from GEO and require low-orbit tracking sensors.

FIGURE 20.12 Computed trajectories of an intercontinental ballistic missile and a depressed-trajectory submarine-launched missile, with a schematic hypersonic glide vehicle for comparison. The inset expands the first 2 500 km.



Illustrative. Space-based infrared sensors provide the first warning, within about a minute of launch.

FIGURE 20.13 Illustrative timeline of a ballistic-missile attack and its detection. Space-based infrared sensors provide the first warning within minutes, long before the missile rises above the radar horizon.

TABLE 20.2 Phases of ballistic-missile flight and the sensors that observe them.

Phase	Duration (ICBM)	Main observables	Primary sensors
Boost	1–5 min	Bright plume (2.7, 4.3 µm bands)	GEO and HEO infrared satellites; LEO tracking satellites
Post-boost / midcourse	20–25 min	Warm warhead, decoys, debris against cold space	Ground- and sea-based radars; LEO infrared tracking with discrimination
Terminal	≈ 1 min	Re-entry heating, radar signature	Terminal-defence radars

20.5.2 From Warning to Defence

Early-warning satellites provide the first **cue** to a missile-defence system. Their tracks tell ground-based radars, such as the US Upgraded Early Warning Radars and the Long Range Discrimination Radar in Alaska, where to look, and they initiate the engagement planning for interceptors. Modern missile-defence architectures increasingly aim for **fire-control-quality** tracks from space — precise enough to launch and guide an interceptor before ground radars can see the target — and for **discrimination** between warheads and decoys.

20.6 Hypersonic Glide Vehicles and the Tracking Layer

Hypersonic glide vehicles (HGVs), now deployed by Russia (Avangard) and China (DF-17), are boosted by rockets. They then glide through the upper atmosphere at 30–60 km altitude and speeds above Mach 5, manoeuvring over thousands of kilometres (Figure 20.12). They challenge existing warning systems in three ways. First, their low altitude keeps them below the horizon of ground radars until late. Second, their glide phase is much dimmer than a rocket plume, although the vehicle is heated by atmospheric friction. Third, their manoeuvres make their target uncertain.

Tracking HGVs requires sensors that are closer and that can follow dim, hot objects against the Earth background through their whole flight. This is the motivation for **proliferated low-Earth-orbit tracking constellations** (Figure 20.14):

- The US Space Development Agency's **Tracking Layer** (Chapter 19) carries wide-field-of-view infrared sensors on satellites at about 1 000 km; Tranche 1 includes 28 tracking satellites, with more in later tranches.
- The Missile Defense Agency's **Hypersonic and Ballistic Tracking Space Sensor (HBTSS)** — two prototypes launched in February 2024 — demonstrates medium-field-of-view sensors that can produce fire-control-quality tracks.
- The US Space Force is also deploying missile-warning and tracking satellites in **medium Earth orbit** ("Epoch 1"), for a layered architecture that is harder to defeat.

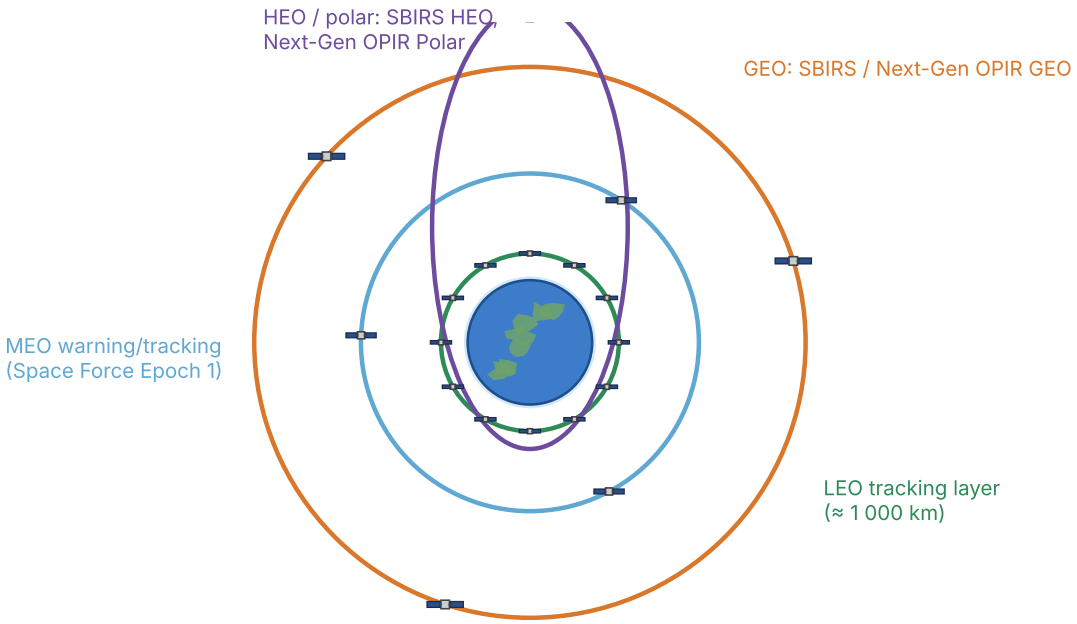


FIGURE 20.14 The layered missile-warning and tracking architecture being built by the USA: geostationary and polar satellites for strategic warning, plus medium- and low-orbit constellations for resilience and tracking.

20.7 Systems in Operation

TABLE 20.3 Space-based missile-warning systems (public information).

Country	System	Orbit	Status (2026)
USA	Defense Support Program (DSP)	GEO	23 launched 1970–2007; last satellites retired or in reserve
USA	SBIRS GEO-1 to GEO-6	GEO	Operational; launched 2011–2022
USA	SBIRS HEO payloads	HEO (hosted)	Operational since 2006–2008; replacement payloads added
USA	Next-Gen OPIR GEO (2 satellites)	GEO	First satellite awaiting launch in late 2026 after delays
USA	Next-Gen OPIR Polar (2 satellites)	HEO	Planned ≈ 2028
USA	SDA Tracking Layer; MDA HBTSS	LEO	Prototypes 2023–2024; Tranche 1 deploying from 2025–26

Country	System	Orbit	Status (2026)
USA	Resilient MW/MT (“Epoch 1”)	MEO	First satellites planned from about 2026–27
Russia	EKS “Kupol” (Tundra satellites)	HEO (Tundra)	About six satellites launched 2015–2022
China	Early-warning satellites (TJS/Huoyan series, reported)	GEO	Operational (limited public information)
France / Europe	SPIRALE (demonstrators, 2009); TWISTER and ODIN’S EYE studies	LEO, GEO	European system under study

20.7.1 SBIRS and Next-Gen OPIR

Each **SBIRS GEO** satellite, built by Lockheed Martin around the A2100 bus, weighs about 4 500 kg at launch. It carries a scanning sensor that covers a wide area continuously and a staring sensor that can be pointed at regions of interest. Both operate in several short- and mid-wave infrared bands. The ground segment processes data from SBIRS, the remaining DSP satellites and hosted sensors together.

The **Next-Generation Overhead Persistent Infrared (Next-Gen OPIR)** programme replaces SBIRS with two GEO satellites (Lockheed Martin, with sensor payloads from RTX and a Northrop Grumman–Ball team) and two polar satellites (Northrop Grumman). Its sensors are designed to be more sensitive to dimmer targets and more resistant to jamming and attack. Like many large defence programmes, it has suffered delays: the first GEO launch, originally planned for 2025, slipped into late 2026.

20.7.2 Other Nations

Russia’s **EKS** (Integrated Space System) uses Tundra satellites in highly elliptical orbits with a “Kupol” (dome) sensor, reportedly able to detect not only ICBM launches but also smaller missiles, and to relay combat commands. China is reported to operate early-warning satellites in GEO and has cooperated with Russia on missile-warning systems. France flew the two **SPIRALE** microsatellites in 2009 to measure infrared backgrounds for a future system. The European Union is funding studies (ODIN’S EYE) and the PESCO **TWISTER** project for a European space-based early-warning capability linked to interceptors.

20.8 Beyond Missiles: Other Infrared Events

The same sensors see many other events. US officials have stated that space-based infrared sensors detected the missile that destroyed Malaysia Airlines flight MH17 over Ukraine in 2014. They have also stated that the sensors gave advance warning to troops in Iraq of the Iranian ballistic-missile attack in January 2020, allowing them to shelter. Large ballistic-missile attacks on Israel in 2024 and on Ukraine throughout the war were detected and tracked from space, cueing the defences. Researchers have also used declassified infrared data to study bolides — small asteroids exploding in the atmosphere — such as the Chelyabinsk event of 2013. US government sensors report such events publicly to NASA’s Center for Near-Earth Object Studies.



FIGURE 20.15 The plume of Starship's sixth flight test seen from the International Space Station on 19 November 2024. Rocket plumes are among the brightest infrared sources on Earth. Image: NASA/JSC (NASA/Don Pettit).

20.8.1 Nuclear Detonation Detection

Nuclear explosions in the atmosphere or space produce a characteristic double-peaked flash of light, bursts of X-rays, gamma rays and neutrons, and an electromagnetic pulse. The **Vela** satellites (1963–1984) carried detectors for these signatures to verify the 1963 Limited Test Ban Treaty. Their successor, the **US Nuclear Detonation Detection System**, uses optical "bhangmeters", X-ray, gamma-ray, neutron and electromagnetic-pulse sensors carried on GPS and DSP/SBIRS satellites. Because GPS satellites are distributed around the Earth, at least four always see any point, allowing a detonation to be located by timing.

20.9 Stability and Arms Control

Reliable early warning is stabilising. If leaders know they will receive unambiguous warning of an attack, they need not adopt hair-trigger postures. They are also less likely to mistake an accident or a space launch for an attack. Conversely, attacks on early-warning satellites could be extremely escalatory, since they might be interpreted as the prelude to a nuclear strike. The **Ballistic Missile Launch Notification Agreement** (1988) between the USA and the Soviet Union, and later agreements, require advance notice of certain launches. Plans for a joint US–Russian data-exchange centre, agreed in 1998, were never implemented. The growing use of the same satellites for both strategic warning and conventional missile defence — "entanglement" — is a subject of concern among arms-control experts.

EXAMPLE: WARNING TIME FOR AN ICBM ATTACK

Using the computed trajectory of Figure 20.12 (burnout at 7.0 km/s and 17° flight-path angle at 200 km altitude):

1. *Boost*: about 3 minutes; the plume becomes visible to a 2.7 μm sensor once it climbs above about 8–10 km, about 40–60 s after launch.
2. *Detection and assessment*: two or more satellites confirm the launch and compute an initial trajectory within about 1–2 minutes of the plume becoming visible.
3. *Flight*: the warhead coasts for about 29 minutes after burnout, reaching an apogee of about 1 040 km and impacting about 10 000 km from the launch site, 32 minutes after launch.
4. *Warning*: subtracting about 3 minutes for detection and reporting leaves roughly 25–28 minutes of warning. A ground-based radar 4 000 km from the target, by contrast, would see the warhead rise above its horizon only about halfway through its flight — giving about half as much warning.



FIGURE 20.16 The exhaust plumes of a Falcon 9 rocket illuminated by sunlight before dawn during the launch of the CRS-35 cargo mission on 13 October 2026, seen from Florida. Twilight launches show the vast extent of rocket plumes at high altitude. Image: NASA/KSC (NASA/John Kraus).

AI IN MISSILE WARNING

Machine learning helps reject clutter and false alarms, classify infrared events from their spectral and temporal signatures, fuse data from many satellites and radars into tracks, and discriminate warheads from decoys. Because errors could have grave consequences, these uses demand rigorous verification and human oversight (Chapters 50 and 53).

KEY TAKEAWAYS

- Space-based infrared sensors detect ballistic-missile launches within about a minute, providing strategic warning and cueing missile defences.
- Rocket plumes emit strongly at 2.7 and 4.3 µm; sensors operating in these atmospheric absorption bands see a dark Earth background and a bright plume once it climbs above the absorbing layer.
- Scanning sensors (DSP) gave way to staring arrays (SBIRS, Next-Gen OPIR); processing rejects clutter from clouds, sun glint and fires and builds three-dimensional tracks from several satellites.
- GEO satellites provide global coverage to about 70° latitude; highly elliptical orbits cover the Arctic.
- ICBMs fly for about 30 minutes, submarine-launched missiles on depressed trajectories for about 10; space-based warning roughly doubles the warning time compared with ground radars alone.
- Hypersonic glide vehicles are driving a shift to proliferated LEO and MEO tracking constellations.
- Early-warning satellites also detect nuclear detonations, bolides and other infrared events; their reliability is central to strategic stability.

REVIEW QUESTIONS

1. Why do missile-warning sensors observe in the 2.7 µm and 4.3 µm absorption bands rather than in atmospheric windows?
2. What angle does the Earth subtend from geostationary orbit? How many 10 µrad pixels would a staring sensor need across that angle?
3. Compare scanning and staring infrared sensors for missile warning.
4. Describe three sources of clutter and how processing rejects them.
5. Why do early-warning constellations combine geostationary satellites with highly elliptical orbits?
6. Outline the phases of an ICBM's flight and how much warning space-based sensors provide in each case for land-based and submarine-launched missiles.
7. Why do hypersonic glide vehicles require a new tracking layer in LEO or MEO?
8. Why is the reliability of early-warning satellites important for strategic stability?

CHAPTER

21 Space Surveillance, Orbital Debris and Space Situational Awareness



Image: NASA/JSC

Some 35 000 catalogued objects — working satellites, dead satellites, spent rocket stages and fragments — orbit the Earth, together with an estimated million pieces of debris too small to track but large enough to destroy a satellite. Space situational awareness (SSA) is the knowledge of where these objects are, where they are going and what they are doing. This chapter uses the public satellite catalogue to show how the orbital population has grown and where it is concentrated, explains the hazards of hypervelocity impacts and the threat of a collisional cascade, and describes the radars, telescopes and analytical methods used to track objects, predict close approaches and decide when to manoeuvre. It then covers debris mitigation rules — from the 25-year guideline to the new five-year rule — active debris removal, re-entry risks and the emerging governance of space traffic.

IN THIS CHAPTER

21.1 A Crowded Environment
21.2 Sources of Debris
21.3 Why Debris Is Dangerous
21.4 Tracking Objects in Space
21.5 Conjunction Assessment and Collision Avoidance
21.6 Debris Mitigation
21.7 Active Debris Removal and Remediation
21.8 Re-entries
21.9 Space Traffic Coordination

21.1 A Crowded Environment

21.1.1 Space Situational Awareness and Space Domain Awareness

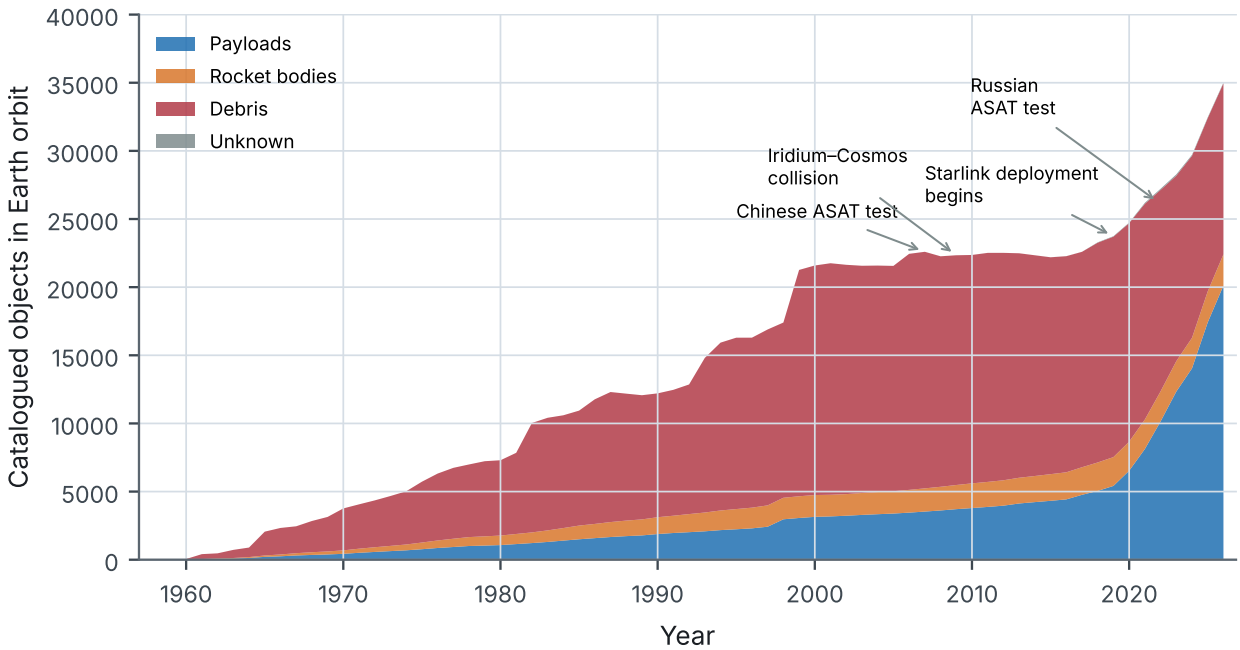
Space situational awareness (SSA) is the knowledge of the population of objects in space and of the space environment needed to operate safely. It covers where each object is, how its orbit is changing, what it is, and whether it threatens another. Military organisations increasingly use the broader term **space domain awareness (SDA)**, which adds understanding of the capabilities and intentions of other actors (Chapter 19). SSA has three main components:

- **Space surveillance and tracking (SST)** of objects in orbit;
- **Space weather** monitoring (Chapters 22 and 23);
- **Near-Earth object** monitoring — asteroids and comets that could hit the Earth (Chapter 41).

This chapter concentrates on the first: the population of artificial objects and the debris problem.

21.1.2 The Numbers

In October 2026 the publicly available catalogue maintained by the US Space Force and redistributed by CelesTrak contained about **35 000** objects in Earth orbit. These were some 20 000 payloads (about 17 000 of them active), 12 500 catalogued debris fragments and 2 300 rocket bodies (Figure 21.1). The total had more than doubled in a decade, driven mostly by the growth of large constellations. Surveillance networks track many more objects than are published — ESA estimates about 40 000 tracked objects — and statistical models estimate far larger populations of smaller debris (Table 21.1).



Computed from launch and decay dates in the CelesTrak SATCAT (publicly catalogued objects only); 35,012 in orbit on 14 Oct 2026.

FIGURE 21.1 Catalogued objects in Earth orbit by type, 1957–2026, computed from the public satellite catalogue. Two destructive events (2007 and 2009) caused sudden jumps; since 2019 large constellations have dominated growth.

TABLE 21.1 Estimated population of debris by size (ESA Space Environment Report 2025, model estimates).

Size	Estimated number	Can it be tracked?	Effect of impact on a satellite
> 10 cm	> 50 000	Yes (most, in LEO)	Catastrophic break-up; satellites manoeuvre to avoid
1–10 cm	≈ 1.2 million	Mostly not	Can destroy a satellite; cannot be avoided or shielded against easily
1 mm – 1 cm	≈ 130 million	No	Damage to sensitive surfaces; penetration of unshielded components
< 1 mm	Many billions	No	Erosion, pitting of windows and solar cells

21.1.3 Where the Objects Are

Objects are not evenly distributed (Figure 21.2). The spatial density in LEO peaks sharply at 450–570 km, where most Starlink satellites fly. It also shows a broader band of old debris and rocket bodies at 700–1 000 km. This is the region of the 2007 Chinese ASAT test, the 2009 Iridium–Cosmos collision and many older break-ups. Debris at these higher altitudes will remain for decades to centuries because the atmosphere is too thin to remove it quickly. Other concentrations are found in the geostationary ring, where about 1 000 objects, active and dead, crowd a narrow belt, and in the MEO navigation orbits.

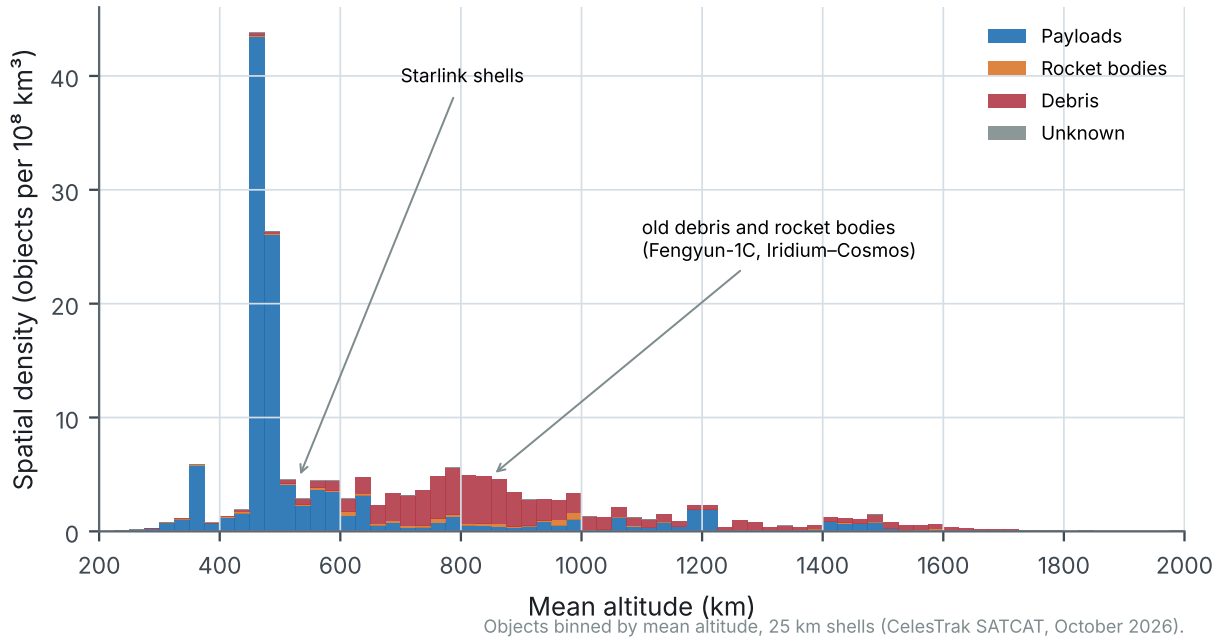


FIGURE 21.2 Spatial density of catalogued objects in low Earth orbit by altitude and type (October 2026).

Plotting every object by altitude and inclination (Figure 21.3) reveals the structure of the population. It shows vertical streaks of satellites in the same orbital planes, the dense band of sun-synchronous orbits at 97–99°, and the shells of Starlink and OneWeb. It also shows clouds of debris spread in altitude around the orbits of the satellites and stages that broke up.

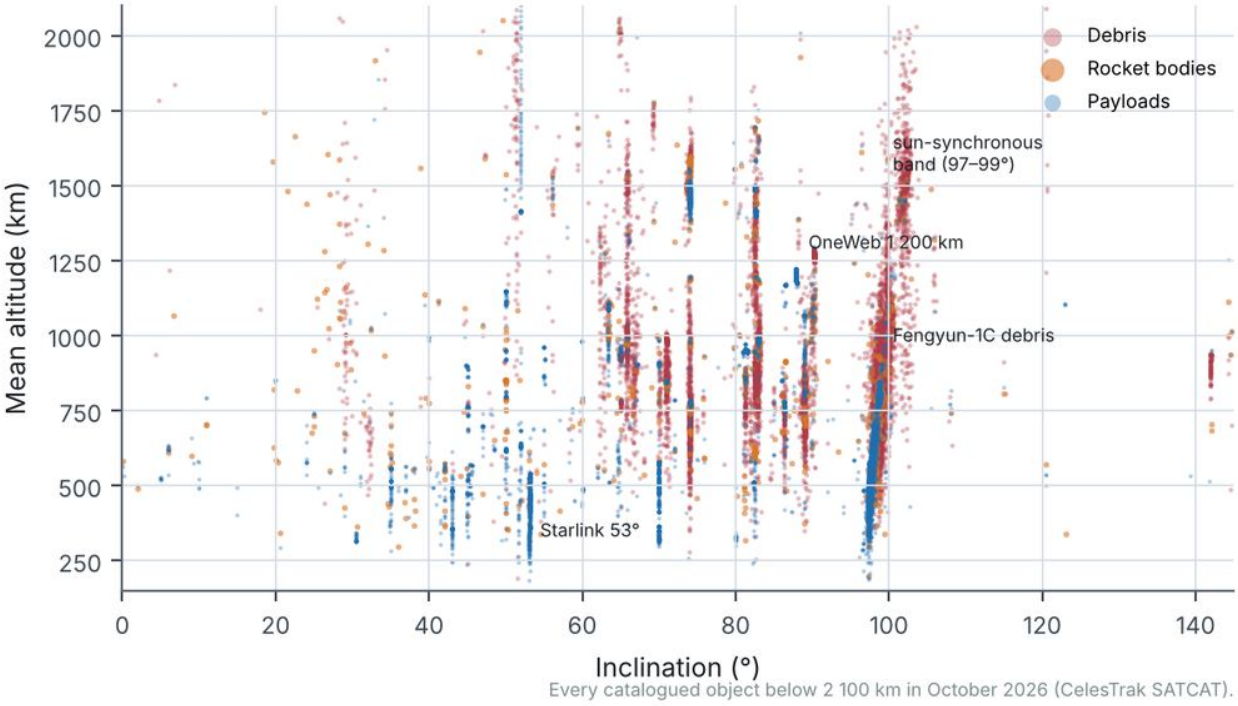


FIGURE 21.3 Altitude and inclination of every catalogued object below 2 100 km in October 2026.

Responsibility for the population is concentrated (Figure 21.4). Objects attributed to the USA — including Starlink — are the most numerous, followed by Russia and China. Russia and China account for most of the catalogued debris, through old explosions, the 2007 and 2021 ASAT tests and recent upper-stage break-ups. Under the Outer Space Treaty and the Liability Convention, the launching state remains responsible for its objects, including debris, indefinitely.

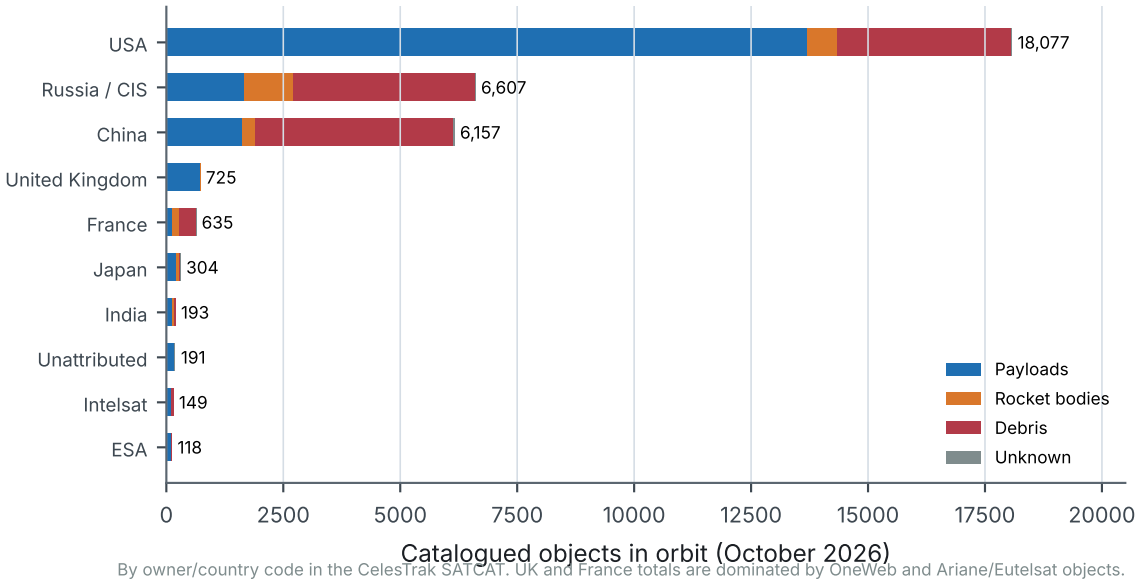


FIGURE 21.4 Catalogued objects in orbit by owner and type (October 2026).

21.2 Sources of Debris

21.2.1 Fragmentations

Most catalogued debris comes from a few hundred **fragmentation events** (Figure 21.5):

- **Explosions** of rocket upper stages and satellites, caused by residual propellant, pressurised tanks or batteries — the most common cause historically. Stages that remain in orbit with leftover propellant can explode decades later, which is why "passivation" (venting propellant and discharging batteries) is now required.
- **Deliberate destruction**, above all the anti-satellite tests of 2007 (Fengyun-1C) and 2021 (Cosmos 1408) (Chapter 19).
- **Collisions**. On 10 February 2009 the active Iridium 33 communications satellite collided with the defunct Russian Cosmos 2251 at about 790 km and a relative speed of 11.7 km/s. It was the first accidental collision between two intact satellites, and it produced more than 2 300 catalogued fragments. Smaller collisions have occurred before and since; in 2021, for example, the Chinese satellite Yunhai 1-02 broke up after being struck by a fragment of debris from a 1996 launch.
- **Anomalous events**: the sudden break-up of old weather satellites (NOAA-16, NOAA-17, DMSP) and, in 2022 and 2024, of Long March 6A upper stages, each producing hundreds of fragments.

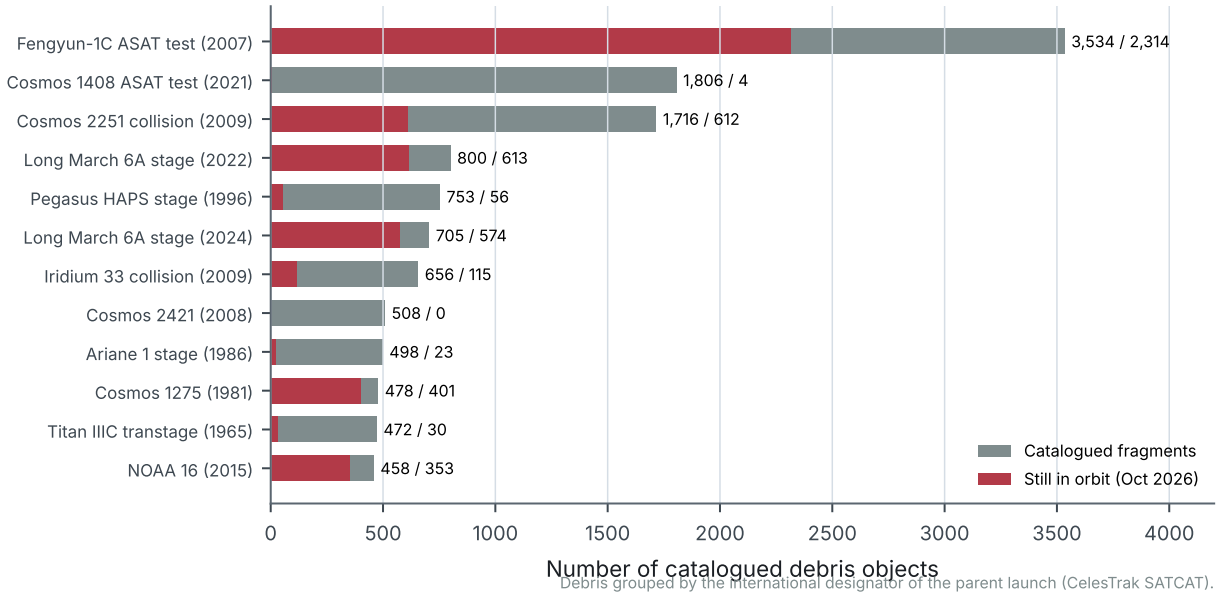


FIGURE 21.5 The largest fragmentation events in the public catalogue: total catalogued fragments and those still in orbit in October 2026.

21.2.2 Other Sources

Smaller debris also comes from **mission-related objects** (lens covers, bolts, adapters released during deployment) and from **solid-rocket-motor** slag and aluminium-oxide dust. Other sources are **paint flakes** and degraded insulation shed by ageing spacecraft (Chapter 5 shows a damaged insulation blanket). There are also droplets of sodium–potassium coolant leaked by Soviet nuclear-powered radar satellites of the 1980s. Natural **micrometeoroids** — dust from comets and asteroids — add a background flux that dominates for sizes below about 0.1 mm.

21.3 Why Debris Is Dangerous

21.3.1 Hypervelocity Impacts

Objects in LEO collide at an average relative speed of about 10 km/s — ten times faster than a rifle bullet. At such speeds, even small particles carry enormous energy (Figure 21.6). A 1 cm aluminium sphere at 10 km/s has a kinetic energy of about 70 kJ, comparable to a small car crash. Both projectile and target are partly vaporised. In practice:

- particles smaller than about 1 mm cause pitting and erosion; windows of the Space Shuttle were replaced after almost every flight because of tiny impacts;
- particles of 1 mm to 1 cm can penetrate unshielded walls, tanks and cables, but **shielding** works against them;
- objects of 1–10 cm are too small to track reliably but too large to shield against — they represent the greatest unmanaged risk;
- objects larger than about 10 cm are tracked, and spacecraft can manoeuvre to avoid them.

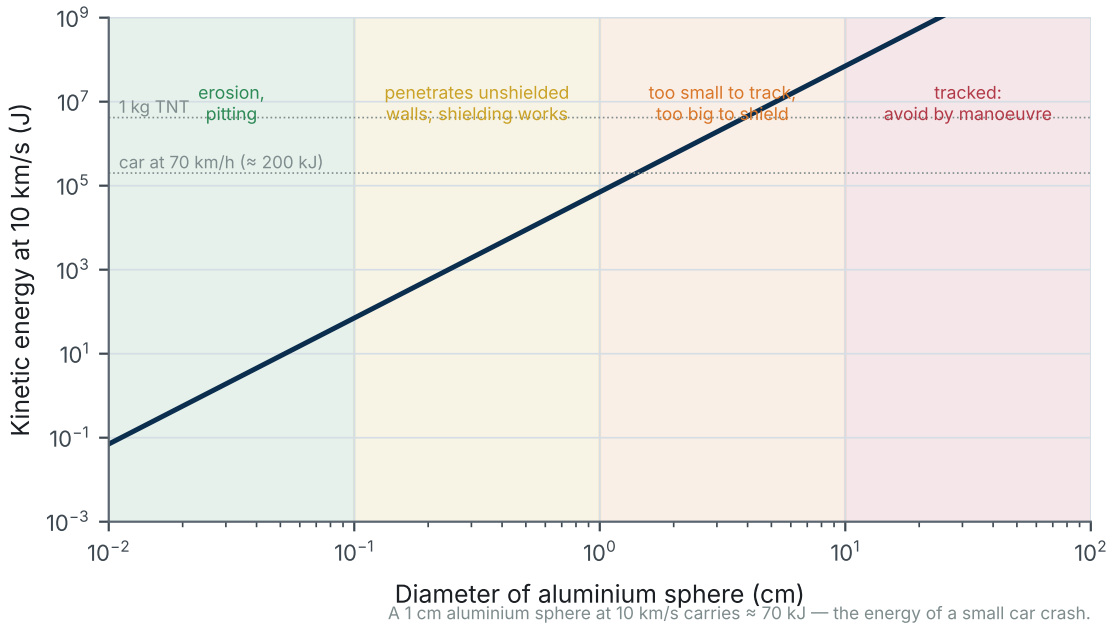


FIGURE 21.6 Kinetic energy of an aluminium sphere at 10 km/s as a function of its diameter, with the effects of impacts in each size range.

21.3.2 Shielding

In 1947, the astronomer Fred Whipple proposed protecting spacecraft from meteoroids with a thin outer "bumper" separated from the main wall. A projectile striking a **Whipple shield** shatters and partly vaporises on the bumper. The resulting cloud of fragments spreads out over the gap and strikes the main wall over a wide area, where it can be stopped. **Stuffed Whipple shields**, with intermediate layers of Kevlar and Nextel ceramic fabric, protect the crewed modules of the ISS against particles up to about 1 cm. Shields are designed and tested with light-gas guns that fire small projectiles at up to about 7–8 km/s (Figure 21.7), complemented by computer simulations. Recovered hardware, such as Hubble's Wide Field and Planetary Camera 2 returned in 2009 (Figure 21.8), provides real data on the impact environment.



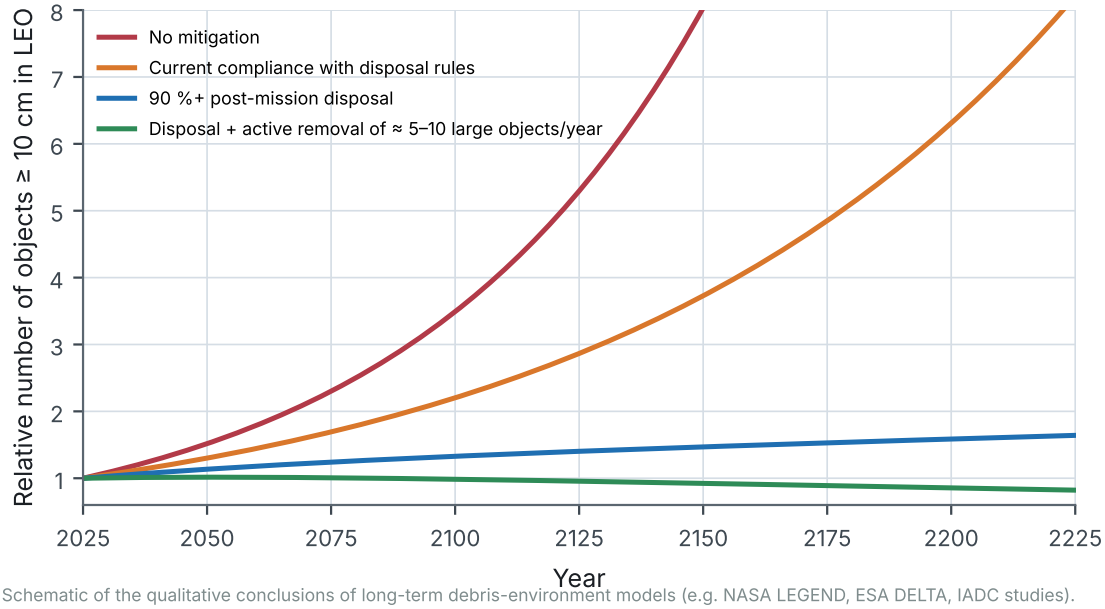
FIGURE 21.7 A target plate after a hypervelocity impact test at 6.8 km/s at NASA's White Sands Test Facility, during tests of materials for the Orion spacecraft. Image: NASA/JSC.



FIGURE 21.8 Engineers of NASA's Orbital Debris Program Office and Hypervelocity Impact Technology team examine impact craters on Hubble's Wide Field and Planetary Camera 2 after its return in 2009, after 16 years in orbit. Image: NASA/JSC.

21.3.3 The Kessler Syndrome

In 1978 Donald Kessler and Burton Cour-Palais of NASA predicted what would happen as the number of objects in orbit grew. Collisions between them would eventually produce debris faster than the atmosphere removes it. This would lead to a self-sustaining **collisional cascade** — now called the **Kessler syndrome**. Long-term models of the debris environment include NASA's LEGEND, ESA's DELTA and the studies of the Inter-Agency Space Debris Coordination Committee. They agree that some LEO altitude bands, especially around 700–1 000 km, may already be above this critical density. Even with no further launches, collisions would slowly increase the number of fragments. The models also show that the outcome depends strongly on behaviour (Figure 21.9). If spacecraft are reliably removed from orbit at the end of their missions, growth can be held in check. Removing a few large derelict objects per year can also stabilise the environment.



Schematic of the qualitative conclusions of long-term debris-environment models (e.g. NASA LEGEND, ESA DELTA, IADC studies).

FIGURE 21.9 Schematic of the qualitative conclusions of long-term debris-environment models under different mitigation scenarios.

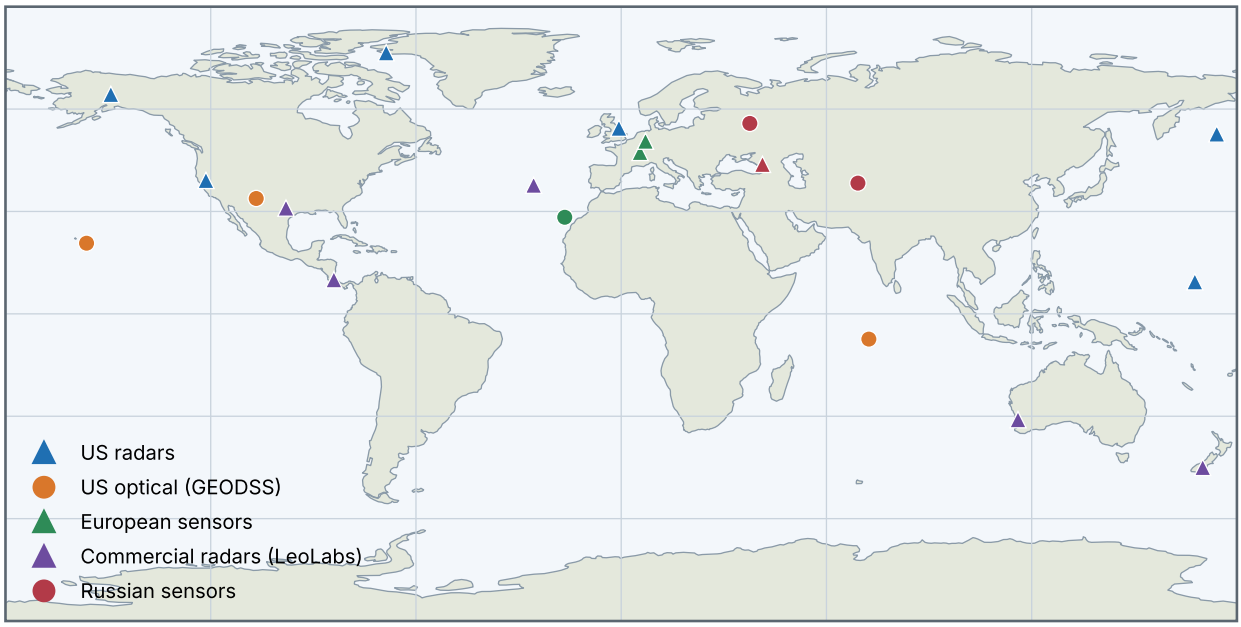
21.4 Tracking Objects in Space

21.4.1 Sensors

Objects in orbit are observed by networks of radars, optical telescopes and, increasingly, sensors in space (Figure 21.10; Table 21.2).

- **Radars** dominate in LEO because they work day and night in all weather and measure range and range-rate directly. Large phased-array radars include the US PAVE PAWS and Upgraded Early Warning Radars (originally built for missile warning, Chapter 20). Others are Cobra Dane in Alaska and the S-band **Space Fence** on Kwajalein Atoll (operational since 2020). Such radars can detect objects down to about 5–10 cm in LEO and track thousands of objects per hour. Tracking and imaging radars such as Germany's TIRA can produce radar images of satellites.

- **Optical telescopes** observe sunlit objects against a dark sky and are best for high orbits, where radar echoes are extremely weak. This is because the radar signal falls as the fourth power of range, but reflected sunlight only as the square (Figure 21.11). The US Ground-based Electro-Optical Deep Space Surveillance (GEODSS) telescopes, commercial networks and European and Russian systems such as ISON monitor the geostationary belt, detecting objects of about 0.5–1 m at GEO.
- **Space-based sensors** avoid weather and day–night limitations. The US Space-Based Space Surveillance satellite (since 2010) observes GEO from LEO. The Geosynchronous Space Situational Awareness Program (GSSAP) satellites — six launched in 2014–2022 and two more in 2026 — patrol near the GEO belt to inspect objects.
- **Commercial providers** now operate their own networks. These include LeoLabs' phased-array radars in several countries and optical networks such as those of ExoAnalytic Solutions and Slingshot Aerospace. Some companies also fly cameras on satellites to observe other objects.



Selected ground-based space-surveillance sensors (approximate locations, public information). ▲ radar, ● optical.

FIGURE 21.10 Selected ground-based space-surveillance sensors around the world (public information).

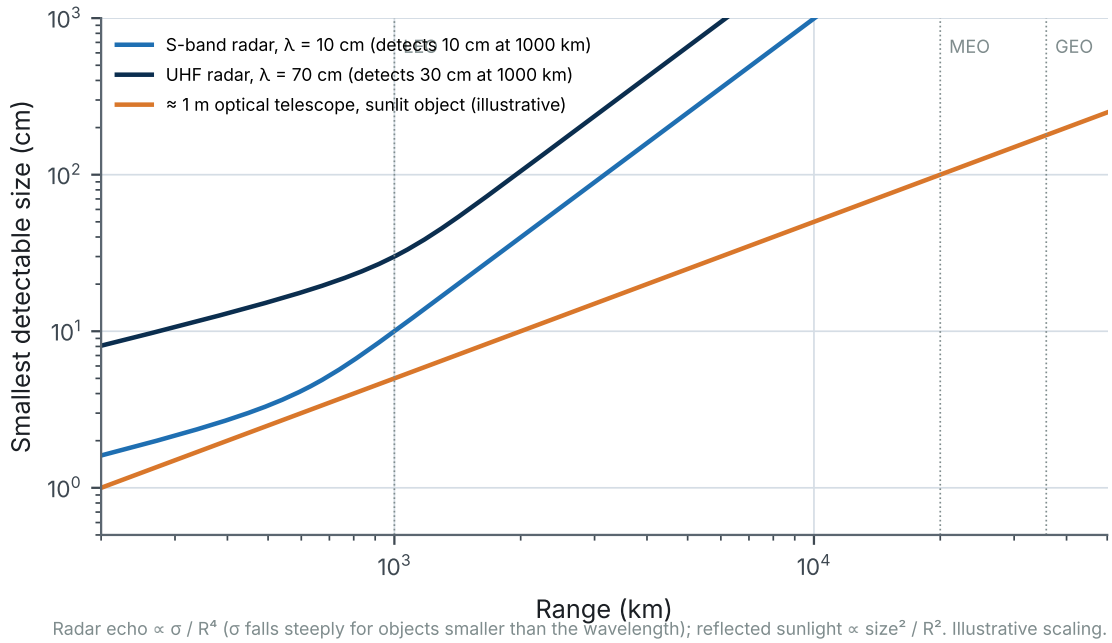


FIGURE 21.11 Approximate smallest detectable object size as a function of range for radars and optical telescopes. Radars excel in LEO; telescopes are needed for GEO.

TABLE 21.2 Selected space-surveillance sensors.

Sensor	Operator	Type	Main role
Space Fence (Kwajalein)	US Space Force	S-band phased-array radar	Uncued detection of small LEO objects
Cobra Dane, PAVE PAWS/UEWR	US Space Force	UHF/L-band phased-array radars	Missile warning and LEO surveillance
GEODSS (Socorro, Maui, Diego Garcia)	US Space Force	1 m optical telescopes	Deep-space (GEO) surveillance
GSSAP, SBSS	US Space Force	Space-based optical	GEO inspection and surveillance
GRAVES, TIRA, EU SST network	France, Germany, EU	Bistatic radar, tracking/imaging radar, optical	European surveillance and tracking
LeoLabs network	LeoLabs (commercial)	UHF/S-band phased-array radars	Commercial LEO tracking and conjunction services
ISON	Russian Academy of Sciences	Optical network	GEO and HEO objects
Okno, Krona	Russia	Optical and radar complexes	Military space surveillance

21.4.2 The Catalogue

The observations are combined into a **catalogue** of orbits. The US Space Force (through its space defence squadrons) maintains the most complete catalogue, publishing data on most unclassified objects through the Space-Track website, where each object has a NORAD catalogue number. Orbits are published as **two-line element sets (TLEs)**, compatible with the SGP4 analytical propagator (Chapter 3). For precise applications, the military also maintains a "special perturbations" catalogue, computed by numerical integration with detailed force models. In addition, operators exchange precise ephemerides. Keeping the catalogue current requires

associating every new observation with the right object. This is difficult after break-ups or when many satellites are deployed together. It also requires modelling atmospheric drag, which changes with solar activity and is the largest source of prediction error in LEO (Chapter 22).

Characterisation goes beyond orbits. Brightness variations (**light curves**) reveal whether an object is tumbling and how fast, and radar cross-sections and radar images reveal its size and shape. Spectroscopy can identify surface materials, and observing manoeuvres reveals whether a satellite is active.

21.5 Conjunction Assessment and Collision Avoidance

21.5.1 Predicting Close Approaches

Every day, screening software compares the predicted trajectories of all catalogued objects to find **conjunctions** — close approaches — within the next several days. For each, the screening centre issues a **conjunction data message (CDM)** to the operators involved, giving the predicted time of closest approach, the miss distance and the uncertainty (covariance) of both objects' positions. Large constellations receive thousands of such messages per week.

The key quantity is the **probability of collision** (P_c), computed in the **encounter plane** perpendicular to the relative velocity (Figure 21.12). LEO encounters last only a fraction of a second at 10–15 km/s, so relative motion can be treated as a straight line. P_c is then the integral of the two-dimensional probability density of the miss vector over a circle. The circle's radius is the combined size of the two objects (Figure 21.13). Counter-intuitively, very uncertain predictions give *low* probabilities, because the probability is spread over a large area ("dilution"); good tracking data are needed to recognise a real threat.

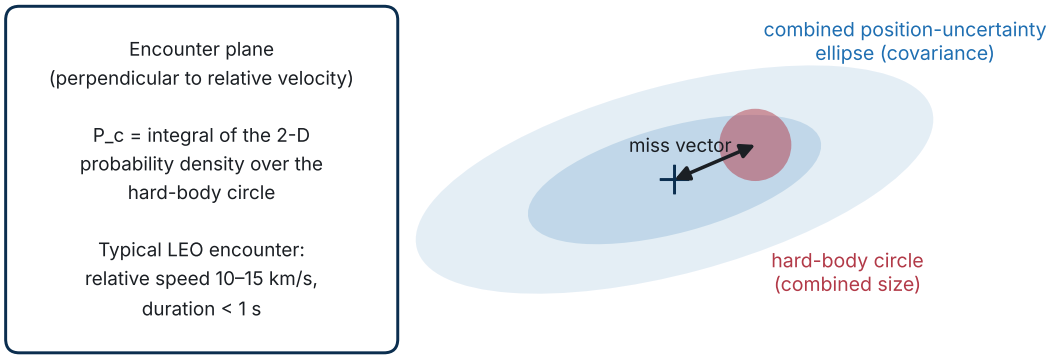


FIGURE 21.12 Geometry of a conjunction in the encounter plane.

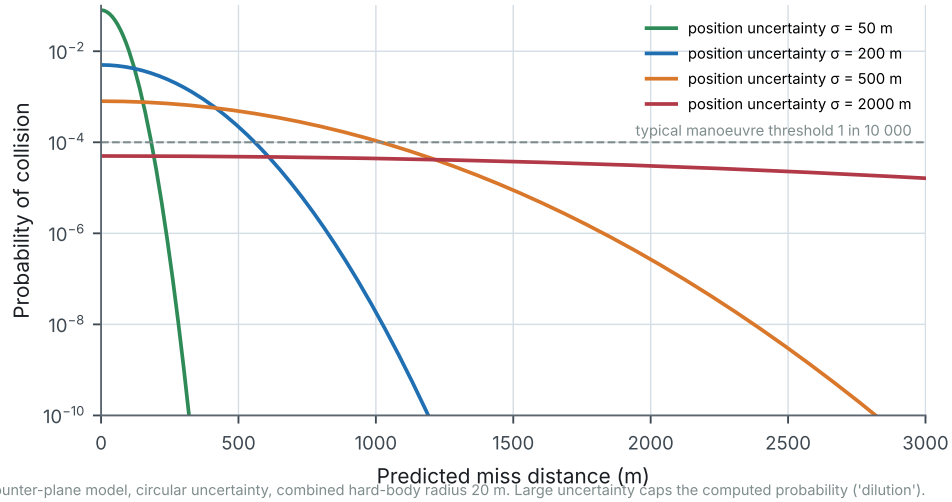


FIGURE 21.13 Probability of collision versus predicted miss distance for different levels of position uncertainty.

21.5.2 Deciding to Manoeuvre

Operators typically plan an avoidance manoeuvre when P_c exceeds about 1 in 10 000 (NASA's threshold for robotic missions), sometimes using more conservative values for crewed vehicles. Each manoeuvre costs propellant, interrupts the mission and may create new conjunctions, so operators weigh the risk carefully; the decision is often made only one or two days before the encounter, when uncertainties have shrunk. The ISS has performed dozens of debris-avoidance manoeuvres since 1999, and its crew has occasionally sheltered in their return vehicles when a threat was detected too late to manoeuvre. Large constellations automate the process: Starlink satellites perform tens of thousands of autonomous avoidance manoeuvres per year using screening data, and share their ephemerides publicly. When both objects can manoeuvre, operators must coordinate so that they do not both move the wrong way.

21.6 Debris Mitigation

21.6.1 Guidelines and Rules

International efforts to limit debris began in the 1990s. The **Inter-Agency Space Debris Coordination Committee (IADC)** was founded in 1993 by the major space agencies. It issued mitigation guidelines in 2002, which were endorsed by the UN Committee on the Peaceful Uses of Outer Space (COPUOS) in 2007. The UN adopted wider **Guidelines for the Long-term Sustainability of Outer Space Activities** in 2019 (Table 21.3). The main principles are:

1. **Limit debris released** during normal operations.
2. **Minimise the potential for break-ups**: passivate spacecraft and stages at the end of life by venting propellant and pressurants and discharging batteries.
3. **Post-mission disposal**: remove spacecraft and stages from the protected LEO region within a limited time — traditionally **25 years**. Do this by lowering their orbits so that drag brings them down, by controlled re-entry, or by boosting them to a disposal orbit. Move GEO satellites at least about 235 km above the geostationary ring (plus a term depending on their area-to-mass ratio) into a "graveyard" orbit.
4. **Prevent on-orbit collisions** through conjunction assessment and manoeuvre capability.

TABLE 21.3 Selected debris-mitigation guidelines and rules.

Instrument	Body	Year	Key provisions
IADC Space Debris Mitigation Guidelines	IADC	2002 (updated)	Passivation, 25-year LEO disposal, GEO graveyard
Space Debris Mitigation Guidelines	UN COPUOS	2007	Endorsed IADC principles for all states
ISO 24113	International Organization for Standardization	2010 (updated)	Engineering requirements for debris mitigation
LTS Guidelines	UN COPUOS	2019	21 guidelines on safety, sustainability and cooperation
Five-year rule	US FCC	adopted 2022, in force 2024	LEO satellites below 2 000 km must be disposed of within 5 years of mission end
Zero Debris approach / Charter	ESA and partners	2023	Aim to stop generating debris by 2030; ≥ 99 % disposal success
French Space Operations Act	France	2008	Legally binding technical regulations, including disposal

The 25-year guideline was set when LEO held a few thousand objects; with tens of thousands of satellites, it is now widely considered too lax. The US Federal Communications Commission's **five-year rule**, in force since 2024 for satellites it licenses, reflects the tightening of expectations. So does ESA's **Zero Debris** approach, which aims to stop creating debris by 2030. Figure 21.14 shows why altitude matters. Satellites below about 500 km re-enter naturally within a few years. Those above about 700 km may stay up for a century or more unless they carry propulsion to lower their orbits.

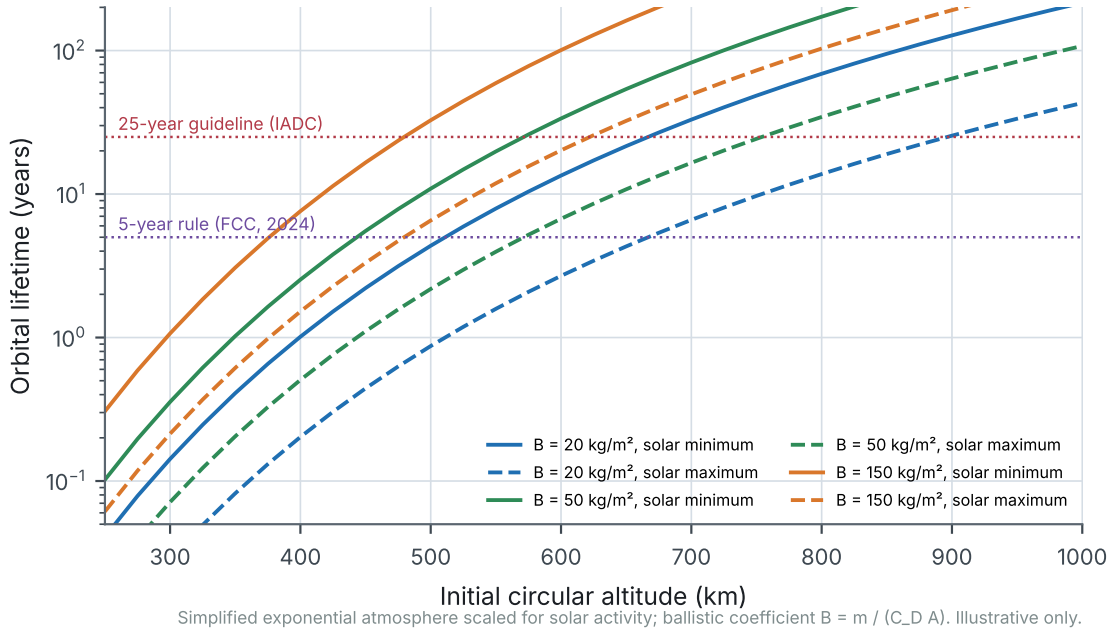


FIGURE 21.14 Approximate orbital lifetime versus initial altitude for different ballistic coefficients and solar activity levels.

21.6.2 Compliance

Compliance with disposal guidelines has improved but remains imperfect. ESA's annual space-environment reports show that most new LEO payloads now comply, largely because so many fly in low orbits that decay quickly. However, many rocket bodies are still abandoned in long-lived orbits, and a few percent of large-constellation satellites fail before they can de-orbit themselves. A failure rate of even 1–2 % in a constellation of 10 000 satellites leaves 100–200 uncontrolled objects.

21.7 Active Debris Removal and Remediation

The most dangerous objects for the long term are the few thousand large, intact derelicts — especially rocket bodies of 1–9 tonnes in crowded orbits. So **active debris removal (ADR)** focuses on capturing them and de-orbiting them. The challenges are considerable: the targets were not designed to be captured, may be tumbling, and have no docking fixtures, and someone must pay for the service.

- Japan's **Astroscale** demonstrated magnetic capture of a cooperative target with **ELSA-d** (2021). With **ADRAS-J** (2024), it approached a discarded H-IIA upper stage to about 15 m and inspected it in detail — the first close approach to a real piece of debris. Its follow-on, **ADRAS-J2**, planned for launch around 2027–28, is designed to capture and remove such a stage.
- ESA's **ClearSpace-1** mission, now planned for around 2028–29, will capture and de-orbit the PROBA-1 satellite (launched in 2001).
- Other concepts include nets, harpoons, robotic arms, drag sails and tethers, ground- or space-based lasers that nudge small debris, and life-extension servicing that keeps valuable satellites working (Chapter 40).

Studies suggest that removing about five to ten of the most dangerous large objects per year, combined with high compliance with disposal rules, could stabilise the LEO environment.

21.8 Re-entries

What goes up comes down. About 2 000 catalogued objects now re-enter each year (Figure 21.15), most of them small objects and constellation satellites that burn up completely. Larger objects pose a risk to people on the ground: dense components such as titanium tanks, steel parts and optics can survive re-entry. Guidelines limit the casualty risk of an uncontrolled re-entry to 1 in 10 000. Larger spacecraft and stages must perform a **controlled re-entry** into a remote ocean area — the South Pacific "spacecraft cemetery" near Point Nemo — or be designed to **demise** completely. Notable uncontrolled re-entries include Skylab (1979), whose debris fell on Western Australia, and several 20-tonne Long March 5B core stages (2020–2022). Another was a fragment of an ISS battery pallet that struck a house in Florida in March 2024, prompting claims for damages under the Liability Convention. Scientists are also studying whether the aluminium oxide produced by thousands of burning satellites each year could affect the upper atmosphere and ozone layer.

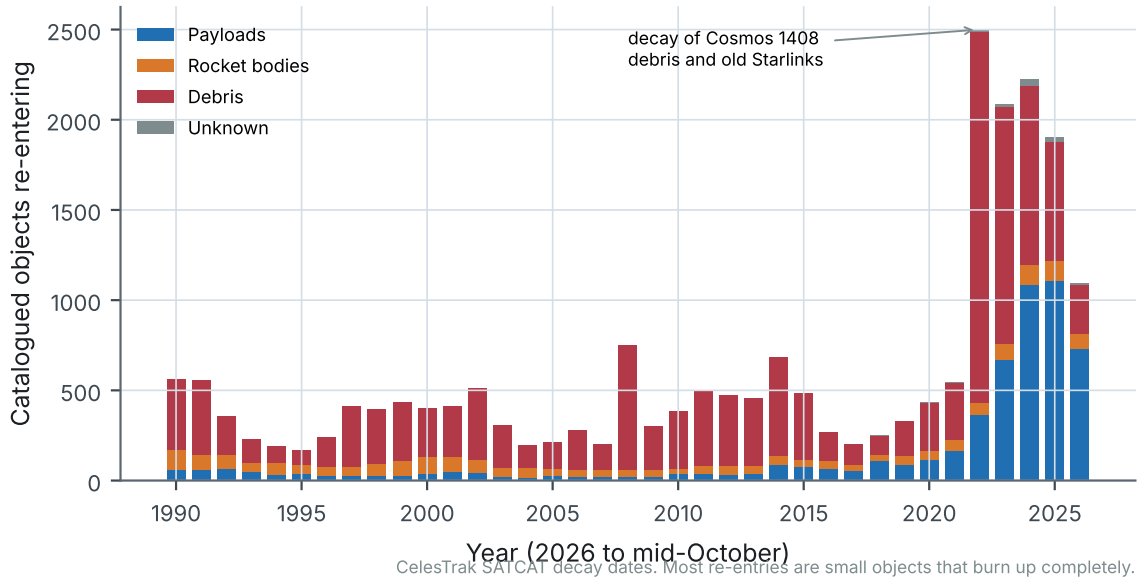


FIGURE 21.15 Catalogued objects re-entering the atmosphere each year by type. Re-entries have increased sharply with large constellations and the decay of debris from the 2021 ASAT test.

21.9 Space Traffic Coordination

There is no international space traffic management system comparable to air traffic control. Each country licenses its own operators, and coordination relies on the sharing of data and good practice. Steps toward more organised **space traffic coordination** include:

- the US Office of Space Commerce's **Traffic Coordination System for Space (TraCSS)**, which is taking over civil conjunction services from the military. It was in pre-production with operators in 2026, though its funding has been contested;
- the **EU Space Surveillance and Tracking** partnership, which provides collision-avoidance services to hundreds of European and international satellites;
- industry initiatives such as the **Space Data Association** and the **Space Sustainability Rating**, which scores missions on their debris-mitigation and data-sharing practices;
- research on automated, interoperable coordination between operators — increasingly using AI (Part XIII).

EXAMPLE: SHOULD WE MANOEUVRE?

A conjunction data message predicts that an Earth-observation satellite will pass a defunct rocket body at a miss distance of 400 m in 36 hours. The combined hard-body radius is 8 m, and the position uncertainty in the encounter plane is about σ = 300 m in each direction.

1. Using the circular-uncertainty approximation,
   $P_c \approx \frac{R^2}{2\sigma^2}\exp\left(-\frac{d^2}{2\sigma^2}\right) = \frac{64}{180\,000}\exp(-0.89) \approx 1.5 \times 10^{-4}$.
2. This exceeds the 1×10^{-4} threshold, so the operator plans a manoeuvre but waits for the next update.
3. Twelve hours later, new tracking reduces σ to 100 m and the predicted miss distance to 150 m:
   $P_c \approx \frac{64}{20\,000}\exp(-1.13) \approx 1.0 \times 10^{-3}$ — a real threat.
4. A small along-track burn of a few centimetres per second, executed half an orbit before the encounter, changes the timing of arrival and increases the miss distance to about 2 km, reducing P_c below 10^{-7}. The operator confirms with the screening service that the new orbit creates no new conjunctions.

AI FOR SPACE SITUATIONAL AWARENESS

Machine learning is used to associate observations with catalogue objects, detect manoeuvres and break-ups, characterise objects from light curves and radar data, improve drag and orbit predictions, prioritise conjunction alerts and recommend avoidance manoeuvres. As constellations grow, automated and AI-assisted coordination between operators will become essential (Chapter 50).

KEY TAKEAWAYS

- About 35 000 objects are catalogued in Earth orbit (October 2026), of which about 17 000 are active payloads; models estimate more than a million debris objects larger than 1 cm.
- Debris comes mainly from explosions, deliberate destruction and collisions; the 2007 Chinese ASAT test and the 2009 Iridium–Cosmos collision are the largest sources still in orbit.
- At 10 km/s, a 1 cm fragment carries the energy of a small car crash; objects of 1–10 cm can neither be tracked reliably nor shielded against.
- Some LEO altitude bands may already exceed the critical density for a collisional cascade (Kessler syndrome); reliable post-mission disposal and removal of large derelicts can stabilise the environment.
- Radars track LEO objects down to about 5–10 cm; telescopes monitor GEO; catalogues are maintained by military, civil and commercial organisations.
- Conjunction assessment uses the probability of collision in the encounter plane; operators typically manoeuvre above about 1 in 10 000.
- Mitigation rules are tightening — from the 25-year guideline to the FCC five-year rule and ESA's Zero Debris approach; active removal missions such as ADRAS-J2 and ClearSpace-1 are preparing to remove debris.

REVIEW QUESTIONS

1. How many objects are catalogued in Earth orbit in 2026, how many are active payloads, and how many debris objects larger than 1 cm are estimated to exist?

2. Calculate the kinetic energy of a 1 cm aluminium sphere (density 2 700 kg/m^3) at 10 km/s, and compare it with a 1-tonne car at 12 m/s.
3. Which events created the largest amounts of debris still in orbit?
4. Why are objects of 1–10 cm in LEO the most dangerous class for operational satellites?
5. What is the Kessler syndrome? Why do studies conclude that removing large derelict objects is more effective than removing small fragments?
6. How is the probability of collision for a conjunction calculated, and what thresholds do operators use to decide on a manoeuvre?
7. Compare the 25-year guideline, the FCC five-year rule and ESA's Zero Debris approach.
8. Describe two active debris-removal techniques and the missions planned to demonstrate them.